

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and six vertices canonically label the seven covering triangles, and an exhaustive classification routes every hypothetical cover to a trace-length, area-loss, or finite equilateral-enclosure contradiction. The body gives the exact geometric and set-theoretic definitions and the compact interfaces used by these three mechanisms. Appendices A–F solve the associated optimization and feasibility problems. The zero-gap nine-point terminal culminates in an exact polynomial positivity certificate over $\mathbb{Q}(\sqrt{3})$; numerical scans are not proof dependencies.

CONTENTS

1. Introduction and Proof Outline	1
2. Common Definitions and Structural Reductions	5
3. Trace-Length Mechanism	8
4. Area-Loss Mechanism	10
5. Finite-Enclosure Obstructions	12
6. Completion of the Proof	22
Appendix A. Structural, Shared-Local, and Signed-Center Optimizations	23
Appendix B. Trace-Length Optimization	29
Appendix C. Area-Loss Optimization	33
Appendix D. Nonzero-Gap Finite-Enclosure Optimization	35
Appendix E. Zero-Gap Nine-Point Optimization	49
Appendix F. Exact Polynomial Positivity Certificate	55

1. INTRODUCTION AND PROOF OUTLINE

1.1. The theorem and the canonical labeling. Set $O := 0 \in \mathbb{R}^2$. With indices in $\mathbb{Z}/6\mathbb{Z}$, choose $V_0 \in \mathbb{R}^2$, let $\text{Rot}_{O,\theta}$ denote ordinary rotation through θ about O , and define

$$\|V_0 - O\| = 1, \quad V_i := \text{Rot}_{O, i\pi/3}(V_0).$$

Put

$$H := \text{conv}\{V_0, \dots, V_5\}, \quad e_{i,i+1} := [V_i, V_{i+1}], \quad r_i := [O, V_i],$$

$$M_i := \frac{O + V_i}{2},$$

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and

$$S := \partial H \cup \bigcup_{i=0}^5 r_i.$$

Then

$$\mathcal{H}^1(\partial H) = 6, \quad \mathcal{H}^1\left(\bigcup_{i=0}^5 r_i\right) = 6, \quad \mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem). *For every $L > 1$, the regular hexagon $H_L := O + L(H - O)$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \quad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle* at V_i . Set

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

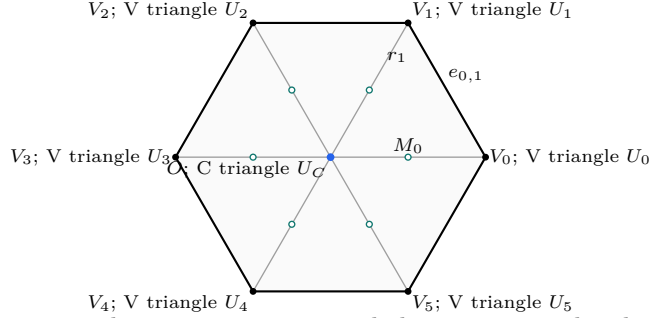


FIGURE 1. The center, vertices, radial segments, and radial midpoints of H .

1.2. Finite classifications. Define the closed C-triangle edge count by

$$N_E(T_C) := \left| \left\{ i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0 \right\} \right|.$$

The closed classifications are

$$T_C \text{ has class CE0} \iff N_E(T_C) = 0,$$

$$T_C \text{ has class CE1} \iff N_E(T_C) = 1,$$

$$T_C \text{ has class CE2} \iff N_E(T_C) = 2.$$

Proposition 2.3 proves that these are the only possibilities.

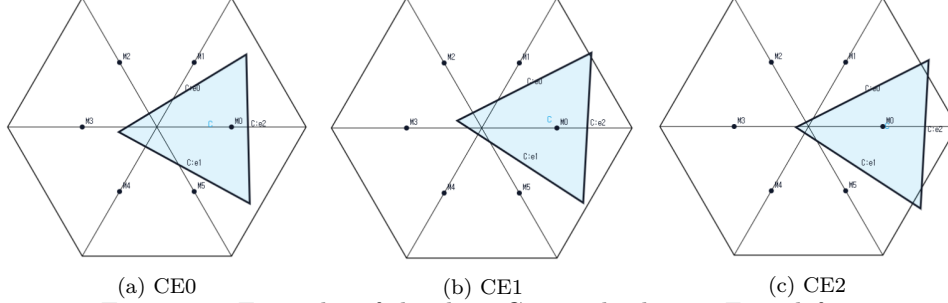


FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H ; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle T_i , put

$$o(T_i) := |\text{vert}(T_i) \setminus H|, \quad n(T_i) := |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|,$$

with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of T_i lie outside H ; it does not say that T_i is disjoint from H , since $V_i \in \text{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in H , still containing V_i in its interior, and with at most two vertices outside H . We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	type
$(1, 0), (2, 0)$	Vd0
$(1, 1)$	Vd1
$(1, 2)$	Vd2
$(2, 1)$	T3-like.

Define the edge and radial parametrizations

$$X_i(s) := V_i + s(V_{i+1} - V_i), \quad R_i(s) := V_i + s(O - V_i) \quad (0 \leq s \leq 1).$$

The actual maximal reaches are defined on the closed V triangle T_i by

$$\begin{aligned} A_i &:= \max\{s \in [0, 1] : V_i + s(V_{i-1} - V_i) \in T_i\}, \\ B_i &:= \max\{s \in [0, 1] : X_i(s) \in T_i\}, \\ C_i &:= \max\{s \in [0, 1] : R_i(s) \in T_i\}. \end{aligned}$$

Equivalently, in terms of the original open V triangle,

$$\begin{aligned} A_i &= \sup\{s \in [0, 1] : V_i + s(V_{i-1} - V_i) \in U_i\}, \\ B_i &= \sup\{s \in [0, 1] : X_i(s) \in U_i\}, \\ C_i &= \sup\{s \in [0, 1] : R_i(s) \in U_i\}. \end{aligned}$$

The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$N_+ := |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|.$$

Lowercase (a_i, b_i, c_i) always denotes selected lower bounds satisfying

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i, \quad 0 \leq c_i \leq C_i;$$

these quantities are never the maxima defining N_+ .

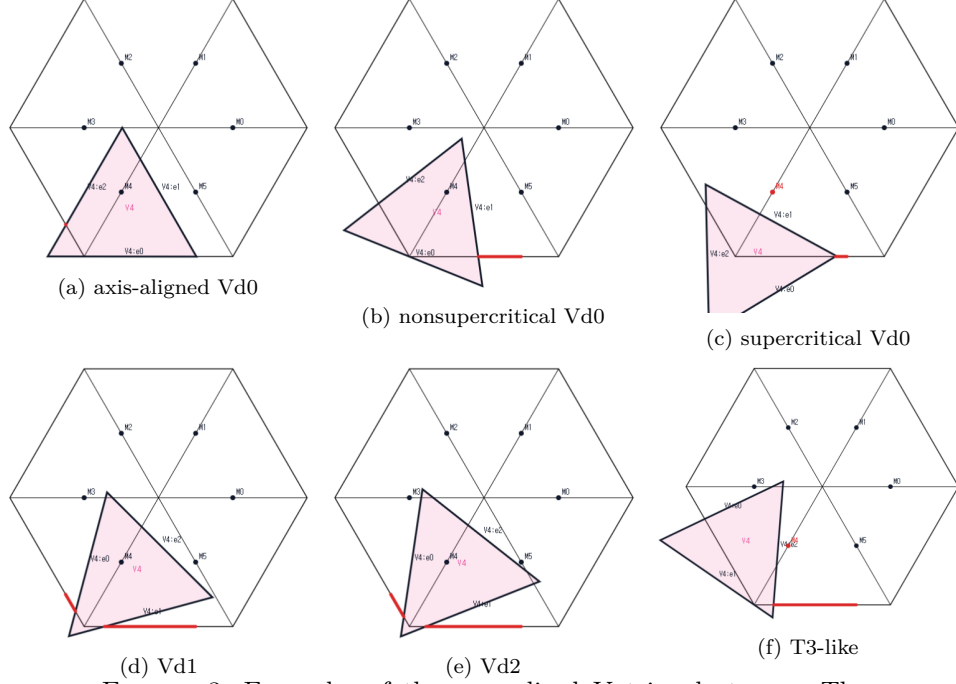


FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \leq 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

The two incident open traces on $e_{i,i+1}$ are

$$U_i \cap e_{i,i+1} = X_i([0, B_i]), \quad U_{i+1} \cap e_{i,i+1} = X_i((1 - A_{i+1}, 1]).$$

Define the boundary-gap set and the number of gap edges by

$$G_i := e_{i,i+1} \setminus ((U_i \cap e_{i,i+1}) \cup (U_{i+1} \cap e_{i,i+1})), \quad N_{\text{gap}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : G_i \neq \emptyset\}|.$$

Thus

$$G_i = \begin{cases} X_i([B_i, 1 - A_{i+1}]), & B_i + A_{i+1} \leq 1, \\ \emptyset, & B_i + A_{i+1} > 1. \end{cases}$$

When equality holds in the first branch, G_i is a singleton and remains uncovered by the two incident open V triangles.

For the case tables, put

$$d := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\}|, \\ t := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) = (2, 1)\}|,$$

and let $N_{\text{sp}} := d + t$. The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\text{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\text{sp}} \geq 3$ is impossible by a trace-length bound on S .

1.3. Exhaustive routing. Lemma 2.9 proves the fundamental equivalence

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Thus Table 1 first dispatches every zero-gap case without reference to the center class. The nonzero-gap part is read from top to bottom; the first applicable row is the unique route for a hypothetical cover. The full classification proof is Proposition 2.15.

N_{gap}	N_+	d	t	$N_E(T_C)$	strategy
0	0	$d + t \leq 6$		$N_E(T_C) \in \{0, 1, 2\}$	1
0	1	$d \geq 1$	$t \geq 0$	$N_E(T_C) \in \{0, 1, 2\}$	1
0	1	$d = 0$	$t \geq 1$	$N_E(T_C) \in \{0, 1, 2\}$	2
0	1	$d = 0$	$t = 0$	$N_E(T_C) \in \{0, 1, 2\}$	3
0	≥ 2	$d + t \leq 6$		$N_E(T_C) \in \{0, 1, 2\}$	2
≥ 1	≥ 2	$d + t \leq 6$		$N_E(T_C) \in \{1, 2\}$	1
≥ 1	0	$d + t \geq 3$		$N_E(T_C) \in \{1, 2\}$	1
≥ 1	1	$d + t \geq 2$		$N_E(T_C) \in \{1, 2\}$	1
≥ 1	0	$d = 0$	$t = 0$	$N_E(T_C) \in \{1, 2\}$	3
≥ 1	0	$d \geq 1, d + t \leq 2$		$N_E(T_C) \in \{1, 2\}$	1
≥ 1	0	$d = 0$	$t \in \{1, 2\}$	$N_E(T_C) \in \{1, 2\}$	3
≥ 1	1	$d = 0$	$t = 0$	$N_E(T_C) \in \{1, 2\}$	3
≥ 1	1	$d = 0$	$t = 1$	$N_E(T_C) \in \{1, 2\}$	3
≥ 1	1	$d = 1$	$t = 0$	$N_E(T_C) = 1$	1
≥ 1	1	$d = 1$	$t = 0$	$N_E(T_C) = 2$	1+3

TABLE 1. The exhaustive routing. Strategy 1 is a trace-length argument, Strategy 2 sums area loss, and Strategy 3 forces a finite residual set into the C triangle and proves an equilateral-enclosure obstruction. Nonzero gap count forces center class CE1 or CE2. In that part the high-count rows are applied first; the final two rows separate the CE1 length route from the detailed CE2 finite-enclosure placement assembly.

1.4. Organization of the proof. Sections 2–5 give the exact definitions and compact interfaces for the common reductions, trace-length, area-loss, and finite-enclosure mechanisms. Appendices A–F solve, respectively, the structural, shared-local, and signed-center optimizations; the trace-length optimization; the area-loss optimization; the nonzero-gap finite-enclosure optimization; the zero-gap nine-point optimization; and the exact polynomial positivity certificate. Section 6 applies the zero-gap matrix before the nonzero-gap rows of Table 1 and closes the proof.

2. COMMON DEFINITIONS AND STRUCTURAL REDUCTIONS

Throughout this section indices lie in $\mathbb{Z}/6\mathbb{Z}$, and we use the open triangles $U_C, U_0, \dots, U_5$, their closures $T_C, T_0, \dots, T_5$, and the maps X_i, R_i fixed in section 1. In particular, all classifications and maximal reaches belong to the closed triangles, whereas boundary coverage and endpoint strictness belong to the original open triangles.

2.1. Open covers and the seven distinguished roles.

Proposition 2.1 (Open, closed, and scaled formulations). *The following assertions are equivalent.*

- (1) H is covered by seven open equilateral triangles of side length 1.
- (2) For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side length $1 - \varepsilon$.
- (3) For some $L > 1$, $H_L := O + L(H - O)$ is covered by seven closed equilateral triangles of side length 1.

Lemma 2.2 (Distinct center and vertex roles). *The covering triangles have a unique labeling for which*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently $O \in \text{int}(T_C)$ and $V_i \in \text{int}(T_i)$.

2.2. Center and vertex classifications.

Recall the exact closed C-triangle count

$$N_E(T_C) = |\{i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0\}|,$$

$$T_C \text{ is CE}k \iff N_E(T_C) = k \quad (k \in \{0, 1, 2\}).$$

A point contact has zero trace length and therefore does not create a second positive trace.

Proposition 2.3 (Exhaustive center classes). *Every C triangle has exactly one of the classes CE0, CE1, CE2; equivalently, $N_E(T_C) \leq 2$. If two edge traces are positive, their edges are adjacent.*

For a closed V triangle put

$$o(T_i) := |\text{vert}(T_i) \setminus H|, \quad n(T_i) := |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|.$$

Lemma 2.4 (Local wedge incidence). *Every original V triangle satisfies $1 \leq o(T_i) \leq 3$. If it has positive-length intersection with one of r_{i-1}, r_{i+1} , at least one of its vertices lies in H ; if it has positive-length intersection with both, at least two of its vertices lie in H .*

Proposition 2.5 (Exact-trace normalization of raw (3,0) roles). *Let U be an original open V_i -triangle, put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a translate $\widehat{U}$ of U , with the same orientation and side length and with closure $\widehat{T}$, such that*

$$V_i \in \widehat{U}, \quad (o(\widehat{T}), n(\widehat{T})) = (2, 0),$$

and

$$T \cap H = \widehat{T} \cap H, \quad U \cap H = \widehat{U} \cap H.$$

Thus all three actual reaches and the inequality $A_i + B_i > 1$ are unchanged. The normalization may be performed simultaneously for every raw (3,0) role.

After this normalization, the exact type dictionary is

$(o(T_i), n(T_i))$	type
$(1, 0), (2, 0)$	Vd0
$(1, 1)$	Vd1
$(1, 2)$	Vd2
$(2, 1)$	T3-like.

Proposition 2.6 (Exhaustive normalized vertex classes). *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, T3-like.*

Proposition 2.7 (T3-like closed-trace majorant). *Let T be the closure of an original T3-like V_i -triangle. There is a translate $\widehat{T}$ of T , with the same orientation and side length, such that*

$$V_i \in \text{relint}(a \text{ side of } \widehat{T}), \quad T \cap H \subsetneq \widehat{T} \cap H,$$

and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$. *Every old open interior point in $H \setminus \{V_i\}$ remains interior to $\widehat{T}$. Since V_i becomes a boundary point, $\widehat{T}$ is only a closed-trace majorant; it is not a replacement open triangle.*

2.3. Actual reaches, open traces, and boundary gaps. The maximal reaches are the closed-triangle maxima

$$A_i = \max\{s \in [0, 1] : X_{i-1}(1-s) \in T_i\},$$

$$B_i = \max\{s \in [0, 1] : X_i(s) \in T_i\},$$

$$C_i = \max\{s \in [0, 1] : R_i(s) \in T_i\},$$

and are equally the corresponding suprema with T_i replaced by U_i . Whenever selected demands occur they satisfy, explicitly,

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i, \quad 0 \leq c_i \leq C_i.$$

In particular,

$$N_+ = |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|$$

is defined only from actual maximal reaches.

Lemma 2.8 (Initial reach segments). *For every normalized V triangle,*

	T_i	U_i
$e_{i-1,i}$	$X_{i-1}([1-A_i, 1])$	$X_{i-1}((1-A_i, 1])$
$e_{i,i+1}$	$X_i([0, B_i])$	$X_i([0, B_i))$
r_i	$R_i([0, C_i])$	$R_i([0, C_i))$

Here each table entry denotes the intersection of the triangle heading its column with the segment heading its row.

The exact gap set is

$$G_i := e_{i,i+1} \setminus ((U_i \cap e_{i,i+1}) \cup (U_{i+1} \cap e_{i,i+1})), \quad N_{\text{gap}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : G_i \neq \emptyset\}|.$$

Lemma 2.9 (Exhaustiveness of the gap split). *For every edge,*

$$G_i = \begin{cases} X_i([B_i, 1-A_{i+1}]), & B_i + A_{i+1} \leq 1, \\ \emptyset, & B_i + A_{i+1} > 1. \end{cases}$$

Equality in the first branch is a singleton gap. Every nonempty G_i lies in a positive-length trace of U_C . Hence CE0, CE1, CE2 permit, respectively, at most 0, 1, 2 gap edges, and under a hypothetical cover

$$N_{\text{gap}} = 0 \iff \partial H \subseteq \bigcup_{i=0}^5 U_i.$$

Lemma 2.10 (T3-like roles are nonsupercritical). *Every original T3-like V triangle satisfies*

$$(1) \quad A_i + B_i \leq 1.$$

In particular, it is not counted by N_+ .

2.4. Strict handoffs and midpoint forcing.

Proposition 2.11 (Strict boundary handoffs). *Assume $\partial H \subseteq \bigcup_{i=0}^5 U_i$. There are simultaneous choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1} < A_i, \quad b_i := \xi_i < B_i,$$

for which $0 < a_i$, $0 < b_i$, and

$$X_{i-1}(1 - a_i), X_i(b_i) \in U_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover, if $N_+ = 1$, every such choice satisfies

$$|\{i : a_i + b_i > 1\}| = 1$$

with the same unique index; if $N_+ \geq 2$, some simultaneous choice satisfies

$$|\{i : a_i + b_i > 1\}| \geq 2.$$

Lemma 2.12 (Self-midpoint obstruction). *For $0 \leq a, b \leq 1$, if a closed unit equilateral triangle contains*

$$\{V_i, X_{i-1}(1 - a), X_i(b), M_i\},$$

then $a + b \leq 1$. Consequently,

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the four displayed points lie in an open unit equilateral triangle, the conclusion is strict.

Proposition 2.13 (CE1/CE2 exactly-one-midpoint property). *If $O \in \text{int}(T_C)$ and $N_E(T_C) \in \{1, 2\}$, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

Remark 2.14. The interior hypothesis is essential: $\text{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1 .

Proposition 2.15 (Exhaustive structural reduction). *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and V triangles used in table 1, and its center class, normalized V classes, actual N_+ , and exact gap sets determine exactly one row of that table.*

3. TRACE-LENGTH MECHANISM

This section gives the exact trace definitions and the compact bounds used by the routing argument. The coordinate optimizations supplying those bounds are solved in Appendix B.

Definition 3.1 (Trace length). For a closed triangle T and a rectifiable set E , define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

If the seven open triangles cover E , their closures satisfy

$$(2) \quad \mathcal{H}^1(E) \leq L_E(T_C) + \sum_{i=0}^5 L_E(T_i).$$

The two targets are the perimeter ∂H and the full skeleton S . Their lengths are 6 and 12, respectively.



∂H : length 6



S : length 12

the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 4. The perimeter and the full skeleton. The skeleton is used only under the hypotheses of the corresponding trace bound.

Theorem 3.2 (Boundary trace table). *For the closures of the original open triangles, the following bounds hold:*

	triangle	hypothesis	$L_{\partial H}$
	T_C	$N_E(T_C) = 0$	0
	T_C	$N_E(T_C) = 1$	$\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
	T_C	$N_E(T_C) = 2$	$< \frac{1}{2}$
(3)	T_i	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}, A_i + B_i \leq 1$	$\leq \frac{1}{2}$
	T_i	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}, A_i + B_i > 1$	$\leq \frac{2}{\sqrt{3}}$
	T_i	$(o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}$	$< \frac{1}{2}$
	T_i	$(o(T_i), n(T_i)) = (2, 1)$	< 1 .

Lemma 3.3 (A positive-support adjacent exceptional triangle is forced). *Suppose $N_E(T_C) \in \{1, 2\}$ and, after symmetry,*

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then there are a supercritical index $s \in \mathbb{Z}/6\mathbb{Z} \setminus \{0\}$ and $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \quad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Theorem 3.4 (Direct CE1/CE2 skeleton count). *If*

$$N_E(T_C) \in \{1, 2\}, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}, \quad N_+ + N_{\text{sp}} \geq 3,$$

then the seven triangles do not cover S .

Proposition 3.5 (Vd2 adjacent-midpoint branch). *Assume*

$$N_E(T_C) = 2, \quad N_+ = 1, \quad \{j : (o(T_j), n(T_j)) \in \{(1, 1), (1, 2)\}\} = \{i\},$$

$$(o(T_i), n(T_i)) = (1, 2), \quad M_{i-1} \in T_i \vee M_{i+1} \in T_i,$$

and

$$\forall j \neq i \quad (o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\}.$$

Then the boundary cannot be covered.

Proposition 3.6 (Branches closed by trace length). *The trace-length mechanism excludes exactly the following routing entries and hybrid subcase:*

- (1) $N_{\text{gap}} = 0$ and $N_+ = 0$;
- (2) $N_{\text{gap}} = 0$, $N_+ = 1$, and $\#\{i : T_i \text{ is Vd1 or Vd2}\} \geq 1$;
- (3) $N_E(T_C) \in \{1, 2\}$ and $N_+ + N_{\text{sp}} \geq 3$;
- (4) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, and $\#\{i : T_i \text{ is Vd1 or Vd2}\} \geq 1$;
- (5)

$$\begin{aligned} N_E(T_C) &= 1, & N_+ &= 1, \\ \#\{i : T_i \text{ is Vd1 or Vd2}\} &= 1, & \#\{i : T_i \text{ is T3-like}\} &= 0; \end{aligned}$$

- (6) the subcase in Proposition 3.5.

4. AREA-LOSS MECHANISM

This section defines the normalized loss exactly and records the local and cyclic interfaces used by the routing argument. Appendix C solves the associated optimizations.

Put

$$A_\Delta := \frac{\sqrt{3}}{4}.$$

Definition 4.1 (Local feasible family and normalized loss). For $(a, b) \in [0, 1]^2$, define

$$\mathcal{T}(a, b) := \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, V_0 + a(V_5 - V_0), V_0 + b(V_1 - V_0)\} \subseteq T \end{array} \right\}.$$

For every closed unit equilateral triangle T , define

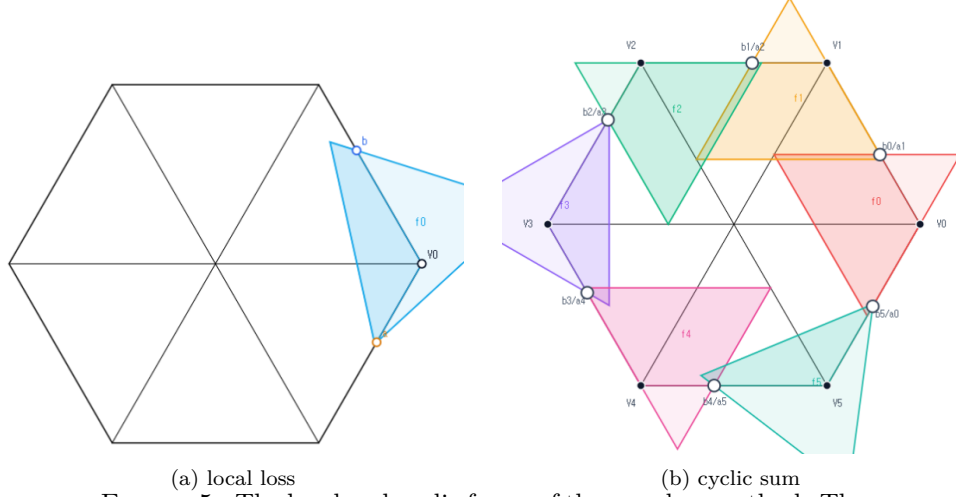
$$\mathcal{L}(T) := \frac{\text{area}(T \setminus H)}{A_\Delta}, \quad \mathcal{L}(a, b) := \inf_{T \in \mathcal{T}(a, b)} \mathcal{L}(T).$$

The corresponding family at V_i is obtained by the hexagon symmetry already fixed in Section 1. In particular, if

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i,$$

then

$$\text{Rot}_{O, -i\pi/3}(T_i) \in \mathcal{T}(a_i, b_i), \quad \mathcal{L}(\text{Rot}_{O, -i\pi/3}(T_i)) = \mathcal{L}(T_i).$$



(a) local loss
 (b) cyclic sum
 FIGURE 5. The local and cyclic forms of the area-loss method. The proof uses the exact variational definition above, not measurements from the picture.

Theorem 4.2 (Unconditional local square loss). *For every $(a, b) \in [0, 1]^2$ with $\mathcal{T}(a, b) \neq \emptyset$,*

$$(4) \quad \mathcal{L}(a, b) \geq \min\{a, b\}^2.$$

If $a + b > 1$, then

$$(5) \quad \mathcal{L}(a, b) \geq \max\{a, b\}^2.$$

Theorem 4.3 (Direct T3-like loss). *Let T_i be an actual T3-like V triangle. Then*

$$(6) \quad A_i + B_i \leq 1.$$

Moreover, if

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i, \quad 0 \leq \mu \leq \frac{1}{2}, \quad a_i, b_i \geq \mu,$$

then

$$(7) \quad \mathcal{L}(T_i) \geq 2\mu - 4\mu^2.$$

Lemma 4.4 (Cyclic area-loss certificate). *Assume the strict handoff data*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1} < A_i, \quad b_i := \xi_i < B_i.$$

Then

$$\#\{i : a_i + b_i > 1\} \geq 2 \implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1,$$

and

$$\begin{aligned} \#\{i : a_i + b_i > 1\} &= 1, \\ \forall i [T_i \text{ is Vd0 or T3-like}], \quad \exists j [T_j \text{ is T3-like}] &\implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1. \end{aligned}$$

Proposition 4.5 (Branches closed by area). *The area-loss mechanism excludes:*

- (1) $N_{\text{gap}} = 0$ and $N_+ \geq 2$;
 (2)

$$\begin{aligned} N_{\text{gap}} &= 0, & N_+ &= 1, \\ \#\{i : T_i \text{ is Vd1 or Vd2}\} &= 0, & \#\{i : T_i \text{ is T3-like}\} &\geq 1. \end{aligned}$$

5. FINITE-ENCLOSURE OBSTRUCTIONS

5.1. Enclosure numbers and local capacities.

Definition 5.1 (Equilateral enclosure number). For $X \in \mathbb{R}^2$ and $r \geq 0$, put

$$\mathbb{B}(X, r) := \{Y \in \mathbb{R}^2 : \|Y - X\| \leq r\}.$$

For every nonempty compact $K \subset \mathbb{R}^2$, put

$$\Lambda(K) := \inf \left\{ \text{side}(Q) : \begin{array}{l} Q \text{ is a closed equilateral triangle,} \\ K \subseteq Q \end{array} \right\}.$$

The support-function evaluation of this infimum is theorem A.9.

Lemma 5.2 (Compact set in an open unit triangle).

$$\left. \begin{array}{l} K \subset \mathbb{R}^2 \text{ is compact,} \\ U \text{ is an open unit equilateral triangle} \end{array} \right\} \implies \Lambda(K) < 1.$$

Normalize at V_0 and use the established radial map R_i .

Definition 5.3 (Anchor hull and radial capacity). For $(a, b, c) \in [0, 1]^3$, define

$$\begin{aligned} K(a, b, c) &:= \text{conv}\{V_0, X_5(1-a), X_0(b), R_0(c)\}, \\ \mathcal{A} &:= \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}. \end{aligned}$$

For

$$0 \leq a, b \leq 1, \quad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

Proposition 5.4 (Projection and symmetry of the anchor capacity). *One has*

$$\{(a, b) \in [0, 1]^2 : \exists c \in [0, 1] ((a, b, c) \in \mathcal{A})\} = \{(a, b) \in [0, 1]^2 : a^2 + ab + b^2 \leq 1\},$$

and the maximum in Definition 5.3 is attained. Moreover,

$$\left. \begin{array}{l} 0 \leq p, q \leq 1, \\ p^2 + pq + q^2 \leq 1 \end{array} \right\} \implies c_{\max}(p, q) = c_{\max}(q, p).$$

Definition 5.5 (Neighboring-radial capacities). For $0 \leq a, b \leq 1$ and $a + b \leq 1$, put

$$C_+(a, b) := \max \left\{ c \in [0, 1] : \begin{array}{l} \exists \Delta [\Delta \text{ is a closed unit equilateral triangle,} \\ \{V_0, X_5(1-a), X_0(b), R_1(c)\} \subseteq \Delta] \end{array} \right\},$$

$$C_-(a, b) := C_+(b, a).$$

The maxima are attained and evaluated in theorem D.1.

Definition 5.6 (Type-aware radial witness). Fix $i \in \mathbb{Z}/6\mathbb{Z}$ and selected reaches satisfying

$$\begin{aligned} 0 &\leq a_j \leq A_j, & 0 &\leq b_j \leq B_j & (0 \leq j \leq 5), \\ a_j^2 + a_j b_j + b_j^2 &\leq 1 & (0 \leq j \leq 5), \\ \mathcal{H}^1(T_{i-1} \cap r_i) &> 0 &\implies a_{i-1} + b_{i-1} \leq 1, \\ \mathcal{H}^1(T_{i+1} \cap r_i) &> 0 &\implies a_{i+1} + b_{i+1} \leq 1. \end{aligned}$$

Define

$$\begin{aligned} \Gamma_i &:= \max \left(\{c_{\max}(a_i, b_i)\} \right. \\ &\quad \cup \{C_+(a_{i-1}, b_{i-1}) : \mathcal{H}^1(T_{i-1} \cap r_i) > 0\} \\ &\quad \left. \cup \{C_-(a_{i+1}, b_{i+1}) : \mathcal{H}^1(T_{i+1} \cap r_i) > 0\} \right), \quad D_i := (1 - \Gamma_i)V_i. \end{aligned}$$

Lemma 5.7 (Type-aware radial forcing). *For the data in Definition 5.6,*

$$D_i \in r_i \subseteq S, \quad D_i \notin \bigcup_{j=0}^5 U_j.$$

Consequently,

$$S \subseteq U_C \cup \bigcup_{j=0}^5 U_j \implies D_i \in U_C.$$

Lemma 5.8 (Common-pair domination). *For $p, q \geq 0$ and $p + q \leq 1$,*

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

5.2. The common nonzero-gap terminals. For $0 \leq p, q \leq 1$ and $p + q \leq 1$, define

$$\begin{aligned} J(p, q) &:= X_0([q, 1 - p]), & c_* &:= c_{\max}(p, q), \\ \mathcal{D}_{(\sqrt{3}/2)(1-c_*)} &:= \mathbb{B}\left(O, \frac{\sqrt{3}}{2}(1 - c_*)\right) = \left\{x : \|x - O\| \leq \frac{\sqrt{3}}{2}(1 - c_*)\right\}. \end{aligned}$$

Theorem 5.9 (Complementary-gap enclosure).

$$\Lambda\left(\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup J(p, q)\right) \geq 1.$$

Proposition 5.10 (One-gap common-enclosure terminal). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_{\text{gap}} = 1, \quad N_+ = 0, \quad N_E(T_C) \in \{1, 2\}, \quad d = 0, \quad t \in \{0, 1, 2\}.$$

After rotation, let

$$\begin{aligned} J &:= G_0 = X_0([\ell, r]), & \ell &:= B_0, & r &:= 1 - A_1, \\ p &:= 1 - r, & q &:= \ell, & c_* &:= c_{\max}(p, q). \end{aligned}$$

Then

$$0 \leq p, q \leq 1, \quad p + q \leq 1.$$

If

$$\forall i \in \mathbb{Z}/6\mathbb{Z} \quad [(p \leq A_i \wedge q \leq B_i) \vee (q \leq A_i \wedge p \leq B_i)],$$

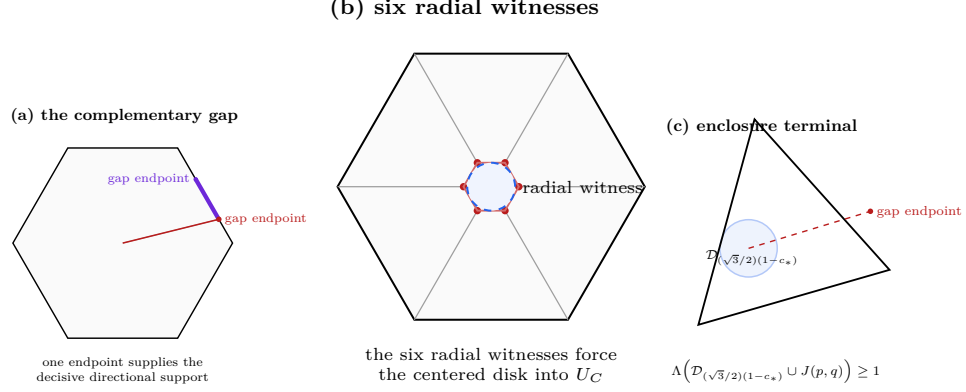


FIGURE 6. The complementary-gap enclosure mechanism. The actual gap $J(p, q)$ and six radial witnesses are center-forced. Their convex hull contains the displayed centered disk, and the enclosure theorem forces side length at least one. The disk is not an additional witness; it is contained in the convex hull of the six finite radial witnesses.

then

$$\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup J \subset U_C, \quad \Lambda(\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup J) \geq 1.$$

In particular, these hypotheses are inconsistent.

Theorem 5.11 (CE2 short-ray obstruction). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_E(T_C) = 2, \quad N_{\text{gap}} = 2, \quad N_+ \in \{0, 1\}, \quad d = 0.$$

Normalize the two positive C -triangle traces to $e_{5,0}$ and $e_{0,1}$, and put

$$p := \min\{s \in [0, 1] : X_5(s) \in T_C\}, \quad q := 1 - \max\{s \in [0, 1] : X_0(s) \in T_C\},$$

$$c_* := c_{\max}(p, q), \quad D_2 := (1 - c_*)V_2, \quad D_4 := (1 - c_*)V_4.$$

Then

$$0 \leq p, q \leq 1, \quad p + q \leq 1.$$

If

$$\forall i \in \{1, 2, 3, 4, 5\} \quad [(p \leq A_i \wedge q \leq B_i) \vee (q \leq A_i \wedge p \leq B_i)],$$

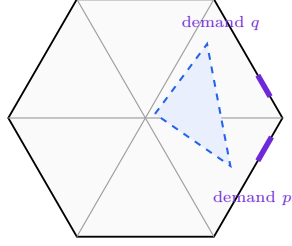
then

$$c_{\max}(q, p) = c_{\max}(p, q) = c_*, \quad D_2, D_4 \in U_C, \quad \{D_2, D_4\} \not\subseteq T_C.$$

In particular, these hypotheses are inconsistent.

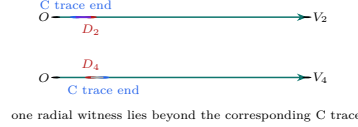
Proposition 5.12 (All nonsupercritical nonzero-gap closures). *Every configuration satisfying the hypotheses of Proposition 5.10 or Theorem 5.11 is impossible.*

(a) two CE2 gaps



p, q are the common boundary demands

(b) transverse center exits



(c) the case split

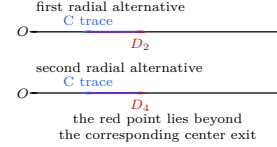


FIGURE 7. The CE2 short-ray terminal. The two boundary gaps determine the common pair (p, q) and the center-forced radial witnesses D_2, D_4 . The theorem proves that at least one of these witnesses lies beyond the corresponding C-triangle trace and is uncovered. The displayed ordering is schematic.

5.3. The transverse seven-point terminal. Suppose

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_{\text{gap}} = 1, \quad N_+ = 1, \quad N_E(T_C) \in \{1, 2\}, \quad (d, t) = (0, 0),$$

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}, \quad \{\sigma\} = \{i : A_i + B_i > 1\}, \quad \sigma = 0.$$

$$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (0 \leq i \leq 5).$$

Put

$$J := G_0 = X_0([\ell, r]), \quad \ell := B_0, \quad r := 1 - A_1,$$

$$P_j := R_j(C_j), \quad K_{\text{tr}} := \{O, M_0, X_0(\ell), X_0(r), P_2, P_3, P_4\}.$$

Lemma 5.13 (The transverse witness is center-forced).

$$K_{\text{tr}} \subset U_C.$$

Proposition 5.14 (CE2 transverse threshold terminal). *No CE2 open unit equilateral triangle contains K_{tr} .*

Proposition 5.15 (CE1 three-transverse return). *No CE1 open unit equilateral triangle contains K_{tr} .*

Proposition 5.16 (Transverse seven-point enclosure).

$$K_{\text{tr}} \subset U_C, \quad \Lambda(K_{\text{tr}}) \geq 1.$$

Consequently the normalized one-gap, one-supercritical all-Vd0 hypotheses above are inconsistent.

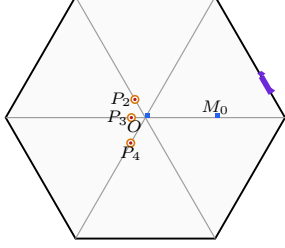
5.4. Supported tails and residual separation. For $0 \leq c < 1/2$, define

$$M_c^{\text{sup}} := \sup\{b \in [0, 1] : \exists a \in [0, 1] ((a, b, c) \in \mathcal{A} \wedge a + b > 1)\},$$

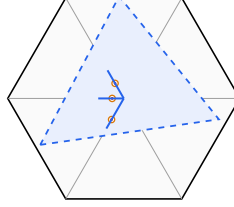
and set

$$M_{1/2}^{\text{sup}} := \frac{1}{2}.$$

The endpoint value is the continuous extension evaluated in theorem D.6, not the supremum of an empty strict feasible set.

(a) the transverse set K_{tr} 

diamonds: $X_0(\ell), X_0(r)$
circles: the only radial endpoints used

(b) the hypothetical C triangle U_C 

P_2, P_3, P_4 are the three
retained radial endpoints

(c) the scalar squeeze

the two gap endpoints
impose an upper demand

the three radial endpoints
force an incompatible return

contradiction

FIGURE 8. The simplified one-gap all-Vd0 enclosure certificate. Only the actual gap endpoints and the three transverse radial endpoints are needed. The gap gives the diameter upper squeeze at the supercritical role; the three radial endpoints supply the CE1/CE2 transverse return. No radial point on r_0 is used. Lengths are schematic.

Proposition 5.17 (T3-like supported-tail terminal). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_{\text{gap}} = 1, \quad N_+ = 1, \quad (d, t) = (0, 1), \quad N_E(T_C) \in \{1, 2\},$$

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}, \quad (o(T_0), n(T_0)) = (2, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \\ (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5).$$

If

$$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1, \quad \varepsilon := 1 - u, \quad C_1 \geq c, \quad M := M_c^{\text{sup}},$$

then

$$R_1(u) = \varepsilon V_1 \in U_C, \quad A_0 + \varepsilon \leq 1, \quad A_0 \leq 1 - M, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M,$$

$$B_1 < M, \quad A_i + B_i \leq 1 \quad (2 \leq i \leq 5),$$

and

$$\sum_{i=2}^5 (A_i + B_i) \geq 4 + M - B_1 > 4.$$

Thus the hypotheses are inconsistent.

Proposition 5.18 (Vd1 supported-tail terminal). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad (d, t) = (1, 0), \quad N_E(T_C) = 2,$$

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}, \quad (o(T_0), n(T_0)) = (1, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \\ (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5).$$

If

$$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1, \quad \varepsilon := 1 - u, \quad C_1 \geq c, \quad M := M_c^{\text{sup}},$$

then

$$R_1(u) = \varepsilon V_1 \in U_C, \quad A_0 + \varepsilon \leq 1, \quad A_0 \leq 1 - M, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M,$$

$$B_1 < M, \quad A_i + B_i \leq 1 \quad (2 \leq i \leq 5),$$

and

$$\sum_{i=2}^5 (A_i + B_i) \geq 4 + M - B_1 > 4.$$

Thus the hypotheses are inconsistent.

Lemma 5.19 (Radial separation interface). *For a routed arm r_i , define*

$$d_C := \max\{x \in [0, 1] : xV_i \in T_C\},$$

$$c_{\text{loc}} := \sup\{s \in [0, 1] : R_i(s) \in U_{i-1} \cup U_i \cup U_{i+1}\}.$$

Then

$$T_C \cap r_i = \{xV_i : 0 \leq x \leq d_C\},$$

$$(U_{i-1} \cup U_i \cup U_{i+1}) \cap r_i \subseteq \{xV_i : 1 - c_{\text{loc}} \leq x \leq 1\}.$$

If $c_{\text{loc}} < 1 - d_C$, then

$$\emptyset \neq \{xV_i : d_C < x < 1 - c_{\text{loc}}\} \subseteq S \setminus \left(U_C \cup \bigcup_{j=0}^5 U_j \right).$$

Definition 5.20 (Boundary residual). For a closed interval $I \subseteq [0, 1]$ and $0 \leq p \leq 1$, put

$$\mathcal{R}_I(p) := 1 - \sup\{x \in [0, 1] : [0, x] \subseteq [0, p] \cup I\}.$$

Proposition 5.21 (Adjacent and nonadjacent Vd separation). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad (d, t) = (1, 0),$$

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$$

$$\{\sigma\} = \{i : A_i + B_i > 1\}, \quad \{\tau\} = \{i : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\},$$

$$\sigma = 0, \quad \tau \in \{1, 2, 3, 4, 5\}, \quad (o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\} \quad (j \neq \tau).$$

If $\tau \in \{1, 5\}$, reflect so that $\tau = 1$ and apply Lemma 5.19 with $i = 2$; then

$$c_{\text{loc}} < 1 - d_C.$$

If $\tau \in \{2, 3, 4\}$, define

$$I_R := \{x \in [0, 1] : X_0(x) \in T_C\}, \quad I_L := \{x \in [0, 1] : X_5(1 - x) \in T_C\},$$

$$\rho_R := \mathcal{R}_{I_R}(B_0), \quad \rho_L := \mathcal{R}_{I_L}(A_0), \quad \rho := \min\{\rho_R, \rho_L\}, \quad D_\tau := \rho V_\tau,$$

$$d_\tau^C := \max\{x \in [0, 1] : xV_\tau \in T_C\}.$$

Then

$$d_\tau^C < \rho < 1 - C_\tau, \quad D_\tau \in S \setminus \left(U_C \cup \bigcup_{j=0}^5 U_j \right).$$

Thus both alternatives contradict the assumed cover of S .

5.5. The two-chart replacement router.

Proposition 5.22 (Two-chart Vd1 replacement and router). *Assume*

$$\begin{aligned} S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \\ (o(T_0), n(T_0)) = (1, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \quad A_i + B_i \leq 1 \quad (i \neq 1), \\ (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5), \quad U_C \cap e_{0,1} = \emptyset, \\ \mathcal{H}^1(T_j \cap r_k) = 0 \quad (2 \leq j \leq 5, k \in \{0, 1\}). \end{aligned}$$

Then

$$\exists \varepsilon > 0 \quad \exists U'_0, U'_1 \quad [U'_k \text{ is an open unit equilateral triangle and } V_k \in U'_k \quad (k = 0, 1)]$$

such that, on setting

$$\begin{aligned} U'_i &:= U_i \quad (2 \leq i \leq 5), \quad T'_i := \overline{U'_i} \quad (0 \leq i \leq 5), \\ G'_i &:= e_{i,i+1} \setminus ((U'_i \cap e_{i,i+1}) \cup (U'_{i+1} \cap e_{i,i+1})), \quad N'_{\text{gap}} := |\{i : G'_i \neq \emptyset\}|, \end{aligned}$$

one has

$$N'_{\text{gap}} \in \{0, 1, 2\}.$$

Define

$$\begin{aligned} A'_i &:= \max\{s \in [0, 1] : V_i + s(V_{i-1} - V_i) \in T'_i\}, \\ B'_i &:= \max\{s \in [0, 1] : V_i + s(V_{i+1} - V_i) \in T'_i\}, \\ C'_i &:= \max\{s \in [0, 1] : R_i(s) \in T'_i\}. \end{aligned}$$

Then

$$\begin{aligned} S \subseteq U_C \cup \bigcup_{i=0}^5 U'_i, \quad N_E(T_C) = 2, \\ (o(T'_i), n(T'_i)) \in \{(1, 0), (2, 0)\} \quad (0 \leq i \leq 5), \quad A'_i + B'_i \leq 1 \quad (0 \leq i \leq 5), \\ A'_0 + B'_0 = A'_1 + B'_1 = 1 - \varepsilon < 1, \\ (A'_i, B'_i, C'_i) = (A_i, B_i, C_i) \quad (2 \leq i \leq 5), \quad |\{i : A'_i + B'_i > 1\}| = 0. \end{aligned}$$

For the recomputed gap rank,

$$\begin{aligned} N'_{\text{gap}} = 0 &\implies S \not\subseteq U_C \cup \bigcup_{i=0}^5 U'_i, \\ N'_{\text{gap}} = 1 &\implies S \not\subseteq U_C \cup \bigcup_{i=0}^5 U'_i, \\ N'_{\text{gap}} = 2 &\implies S \not\subseteq U_C \cup \bigcup_{i=0}^5 U'_i. \end{aligned}$$

The three implications use, respectively, Proposition 3.6, Theorem 5.9, and Theorem 5.11.

Proposition 5.23 (Direct one-Vd placement assembly). *If*

$$N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad d = 1,$$

then

$$S \not\subseteq U_C \cup \bigcup_{i=0}^5 U_i.$$

5.6. **Finite-enclosure register.** Rows A–E below share the exact hypothesis

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i;$$

row F retains the stronger hypothetical cover of H . All entries in the subcase-identifying columns are formulas. In row F, a, b and $K_{\text{wit}}(a, b)$ are the exact objects defined in equations (9) and (13) in the following zero-gap subsection.

Table 2: Exact finite-enclosure register.

	rank data	normalized predicates	exact conclusion
A		$N_E(T_C) \in \{1, 2\}, \quad G_0 = X_0([\ell, r]),$	
	$N_{\text{gap}} = 1$	$\ell = B_0, \quad r = 1 - A_1, \quad p = 1 - r, \quad q = \ell,$	$\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup G_0$
	$N_+ = 0$	$0 \leq p, q \leq 1, \quad p + q \leq 1, \quad c_* = c_{\max}(p, q),$	$\subset U_C,$
	$d = 0$	$\forall i \in \mathbb{Z}/6\mathbb{Z},$	$\Lambda(\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup G_0) \geq 1$
B		$(p \leq A_i \wedge q \leq B_i)$	
		$\vee (q \leq A_i \wedge p \leq B_i)$	
	$N_{\text{gap}} = 2$	$N_E(T_C) = 2, \quad p = \min\{s \in [0, 1] : X_5(s) \in T_C\},$	
	$N_+ \in \{0, 1\}$	$q = 1 - \max\{s \in [0, 1] : X_0(s) \in T_C\},$	$c_{\max}(q, p) = c_*,$
C	$d = 0$	$0 \leq p, q \leq 1, \quad p + q \leq 1,$	$D_2, D_4 \in U_C,$
		$c_* = c_{\max}(p, q), \quad D_j = (1 - c_*)V_j \quad (j \in \{2, 4\}),$	$\{D_2, D_4\} \not\subseteq T_C$
		$\forall i \in \{1, \dots, 5\},$	
		$(p \leq A_i \wedge q \leq B_i)$	
D ₁	$N_{\text{gap}} = 1$	$N_E(T_C) \in \{1, 2\}, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	
	$N_+ = 1$	$\{i : A_i + B_i > 1\} = \{0\},$	$K_{\text{tr}} \subset U_C \wedge \Lambda(K_{\text{tr}}) \geq 1$
	$(d, t) = (0, 0)$	$\forall i \in \mathbb{Z}/6\mathbb{Z} \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}$	
		$N_E(T_C) \in \{1, 2\}, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	
D ₂		$(o(T_0), n(T_0)) = (2, 1),$	
	$N_{\text{gap}} = 1$	$M_1 \in T_0, \quad A_1 + B_1 > 1,$	
	$N_+ = 1$	$\forall i \in \{1, \dots, 5\} \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\},$	$\sum_{i=2}^5 (A_i + B_i) \geq 4 + M -$
	$(d, t) = (0, 1)$	$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1,$	$B_1 > 4$
E ₁		$\varepsilon = 1 - u, \quad C_1 \geq c, \quad M = M_c^{\text{sup}}$	
		$N_E(T_C) = 2, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	
	$N_{\text{gap}} \in \{1, 2\}$	$(o(T_0), n(T_0)) = (1, 1),$	
	$N_+ = 1$	$M_1 \in T_0, \quad A_1 + B_1 > 1,$	$\sum_{i=2}^5 (A_i + B_i) \geq 4 + M -$
E ₂		$\forall i \in \{1, \dots, 5\} \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\},$	$B_1 > 4$
	$(d, t) = (1, 0)$	$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1,$	
		$\varepsilon = 1 - u, \quad C_1 \geq c, \quad M = M_c^{\text{sup}}$	
		$N_E(T_C) = 2, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	
E ₃		$\{\sigma\} = \{i : A_i + B_i > 1\},$	
	$N_{\text{gap}} \in \{1, 2\}$	$\{\tau\} = \left\{ i : \begin{smallmatrix} (o(T_i), n(T_i)) \in \\ \{(1, 1), (1, 2)\} \end{smallmatrix} \right\},$	
	$N_+ = 1$	$\sigma = 0, \quad \tau \in \{1, 5\},$	$c_{\text{loc}} < 1 - d_C$
	$(d, t) = (1, 0)$	$\forall j \neq \tau,$	
		$(o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\}$	

rank data	normalized predicates	exact conclusion
	$N_E(T_C) = 2, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	
	$\{\sigma\} = \{i : A_i + B_i > 1\},$	$d_\tau^C < \rho < 1 - C_\tau,$
$N_{\text{gap}} \in \{1, 2\}$	$\{\tau\} = \left\{ i : \begin{smallmatrix} (o(T_i), n(T_i)) \in \\ \{(1,1), (1,2)\} \end{smallmatrix} \right\},$	$D_\tau \in S \setminus (U_C \cup \bigcup_{j=0}^5 U_j)$
$N_+ = 1$	$\sigma = 0, \quad \tau \in \{2, 3, 4\},$	
$(d, t) = (1, 0)$	$\forall j \neq \tau,$	
	$(o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\}$	
$N_{\text{gap}} = 0$		
$N_+ = 1$	$H \subseteq U_C \cup \bigcup_{i=0}^5 U_i$	$K_{\text{wit}}(a, b) \subset U_C \wedge$
$(d, t) = (0, 0)$		$\Lambda(K_{\text{wit}}(a, b)) \geq 1$

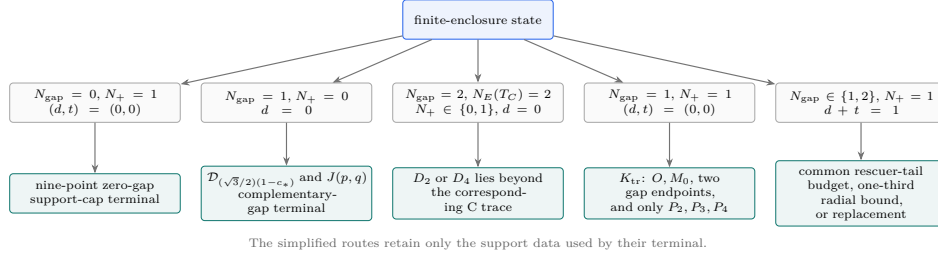


FIGURE 9. Roadmap of the finite-enclosure mechanism. The one-gap all-Vd0 witness uses only three transverse radial endpoints; the two-gap terminal uses one of the two displayed radial witnesses; and the T3-like and Vd1 paths share one rescuer-tail lemma.

Proposition 5.24 (All nonzero-gap finite-enclosure rows). *Every nonzero-gap row of Table 1 assigned to the finite-enclosure strategy is impossible.*

5.7. Zero-Gap Nine-Point Obstruction. Under a hypothetical cover, the zero-gap branch considered here is

$$N_{\text{gap}} = 0, \quad N_+ = 1, \quad (d, t) = (0, 0).$$

Rotate the unique supercritical role to index 4. Then

$$(8) \quad \xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4,$$

and, on putting

$$(9) \quad a := 1 - \xi_3, \quad b := \xi_4, \quad p := 1 - b, \quad q := 1 - a,$$

one has

$$(10) \quad \begin{aligned} 0 < a, b < 1, \quad a = a_4 < A_4, \quad b = b_4 < B_4, \\ a + b > 1, \quad a^2 + ab + b^2 < 1, \quad a_i \geq p, \quad b_i \geq q \quad (0 \leq i \leq 5). \end{aligned}$$

These relations are supplied by theorem E.2.

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \quad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a, b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Using the previously defined radial envelope $c_{\max}$, put

$$c_* := c_{\max}(p, q), \quad h := \frac{\sqrt{3}}{2}, \quad r_* := h(1 - c_*),$$

$$P_i^{\text{rad}} := (1 - c_*)V_i \quad (0 \leq i \leq 5), \quad \mathbb{B}(0, r_*) := \{X : \|X\| \leq r_*\}.$$

The direct radial forcing lemma gives

$$(11) \quad \mathbb{B}(0, r_*) \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_C.$$

Indeed, this is precisely theorem E.3.

For the asymmetric witnesses, define the two adjacent-handoff circles by

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \quad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1 - a), 1).$$

The corresponding maximal nondegenerate line pieces of the interior frontier are defined intrinsically by

$$\{\ell_-\} := \left\{ [X, Y] : \begin{array}{l} X, Y \in \mathbb{R}^2, \quad X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_- \neq \emptyset, \\ \nexists X', Y' \in \mathbb{R}^2 \quad [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\},$$

$$\{\ell_+\} := \left\{ [X, Y] : \begin{array}{l} X, Y \in \mathbb{R}^2, \quad X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_+ \neq \emptyset, \\ \nexists X', Y' \in \mathbb{R}^2 \quad [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\}.$$

Finally, define the three frontier points by

$$\{Q_0\} := \ell_- \cap \ell_+, \quad \{Q_-\} := \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} := \ell_+ \cap \mathcal{C}_+.$$

Their existence and uniqueness follow from theorems E.1 and E.4; their direct exclusion from all six V triangles is theorem E.5. Thus

$$(12) \quad Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i, \quad Q_-, Q_0, Q_+ \in U_C.$$

The public compact witness is

$$(13) \quad K_{\text{wit}}(a, b) := \mathbb{B}(0, r_*) \cup \{Q_-, Q_0, Q_+\},$$

$$K_{\text{wit}}(a, b) \subset U_C.$$

For the schematic support picture, for $X \in \mathbb{R}^2$ and $r \geq 0$ write

$$\text{Cap}(X, r) := \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

appendix E constructs the labeled Newton inner points and proves the four exact cap overlaps depicted below.

Theorem 5.25 (Witness enclosure). *For every pair (a, b) satisfying (10),*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

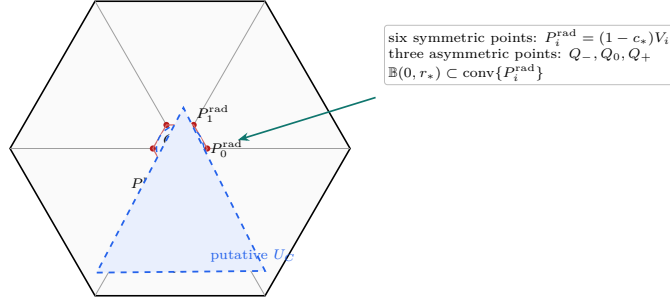
Theorem 5.26 (Zero-gap obstruction). *There is no cover of H by open unit equilateral triangles $U_C, U_0, \dots, U_5$ for which*

$$N_{\text{gap}} = 0, \quad N_+ = 1, \quad (d, t) = (0, 0).$$

Proposition 5.27 (Cases excluded by the nine-point obstruction). *Every branch satisfying*

$$N_{\text{gap}} = 0, \quad N_+ = 1, \quad (d, t) = (0, 0)$$

is impossible, independently of the center class.



$$K_{\text{wit}} = \mathbb{B}(0, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C \text{ under a hypothetical cover}$$

FIGURE 10. The zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB -frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

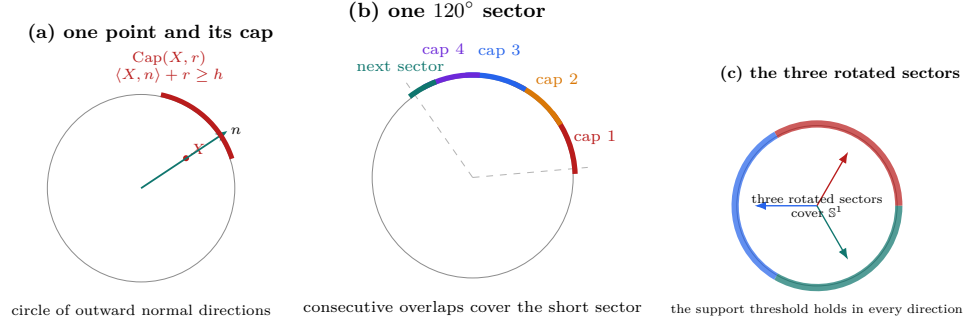


FIGURE 11. The support-cap conclusion for the zero-gap witness. A point and a disk radius determine an arc of normal directions in which their support is at least h . Four consecutive overlap checks cover one sector, and its three 120° rotations cover the full normal circle. The arc widths are schematic; the cyclic order and overlap pattern are the exact hypotheses of the support-arc lemma.

6. COMPLETION OF THE PROOF

Corollary 1.2 follows from Proposition 2.1.

APPENDIX A. STRUCTURAL, SHARED-LOCAL, AND SIGNED-CENTER OPTIMIZATIONS

For all coordinate evaluations in this appendix, choose the global Euclidean chart

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right).$$

Every conclusion is invariant under the hexagon symmetries already fixed in the body.

A.1. Module A.1: open–closed shrinking feasibility.

Exact interface. The three cover formulations are those of theorem 2.1.

Variables and feasible set. Let $U_1, \dots, U_7$ be open unit equilateral triangles covering the compact hexagon H . For $p \in H$ define

$$m_j(p) := \text{dist}(p, \mathbb{R}^2 \setminus U_j), \quad m(p) := \max_{1 \leq j \leq 7} m_j(p).$$

Objective. Find a single inward displacement that may be applied to all twenty-one side lines while retaining a cover of H .

Exact result.

Lemma A.1 (Uniform open–closed shrinking). *There is $\rho > 0$ such that moving every side of every U_j inward through distance ρ produces seven closed equilateral triangles covering H , each of side length $1 - 2\sqrt{3}\rho > 0$.*

Solution.

Body consequence. This is the quantitative implication used in theorem 2.1.

A.2. Module A.2: center and V-triangle incidence feasibility.

Exact interface. The C-triangle count $N_E(T_C)$ and the V-triangle incidence pair $(o(T_i), n(T_i))$ are defined in sections 1 and 2.

Variables and feasible set. For the center calculation take relative-interior parameters $0 < s, t < 1$ on two nonadjacent edges. For the V calculation normalize to $i = 0$ and use the affine wedge chart

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0),$$

with V_0 as the chart origin. Its squared Euclidean norm is

$$(14) \quad \|(x, y)\|^2 = x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x - y)^2 + xy.$$

Within the closed unit ball about V_0 , the part of H is exactly

$$W := \{(x, y) : x \geq 0, y \geq 0\},$$

and the adjacent radial segments are

$$r_1 = \{(1, y) : 0 \leq y \leq 1\}, \quad r_5 = \{(x, 1) : 0 \leq x \leq 1\}.$$

Objective. Determine the maximum number of positive C-triangle edge incidences and the minimum number of V-triangle vertices forced into the local wedge by one or two positive radial incidences.

Exact result.

Lemma A.2 (Center and V-triangle incidence calculation). *A closed unit equilateral triangle with O in its interior has positive trace on at most two edges of H , and any two such edges are adjacent. An original V triangle has $1 \leq o(T_i) \leq 3$; one positive adjacent radial trace forces at least one of its vertices into H , and two such traces force at least two vertices into H .*

Solution.

Body consequence. The first conclusion supplies theorem 2.3; the remaining conclusions supply theorem 2.4 and the finite vertex-type list.

A.3. Module A.3: exact-trace normalization and interval geometry.

Exact interface. The input is an original V triangle U_i , its closure T_i , the incidence pair $(o(T_i), n(T_i))$, and the actual reaches A_i, B_i, C_i defined in section 1. The output must preserve the exact open and closed traces whenever a role is normalized.

Variables and feasible set. Normalize to $i = 0$ in the wedge chart of Module A.2. Put

$$D_t := 1 - t + t^2, \quad z := \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

After relabeling the triangle vertices, every orientation is in one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

For Type I set

$$(15) \quad U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\},$$

and for Type II set

$$(16) \quad U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

In both cases

$$(17) \quad V_0 \in \text{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z.$$

The Type-II vertex in the wedge is

$$(18) \quad Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right),$$

whose coordinates are positive on the feasible set.

Objective. For a raw $(3, 0)$ role, find a same-orientation translation with identical trace in H and type $(2, 0)$. For a T3-like role, find the exact closed-trace majorant and determine its actual boundary-reach sum. Finally, determine all open and closed initial reach intervals and their exact gap set.

Exact result.

Lemma A.3 (Exact-trace translations and initial intervals). *The following hold.*

- (1) *Every raw $(3, 0)$ role admits a same-orientation translate $\widehat{U}$ for which*

$$V_i \in \widehat{U}, \quad (o(\widehat{U}), n(\widehat{U})) = (2, 0), \quad U \cap H = \widehat{U} \cap H, \quad \overline{U} \cap H = \widehat{\overline{U}} \cap H.$$

- (2) *Every T3-like role has Type II orientation and admits the closed-trace majorant of theorem 2.7. Its exact reaches satisfy $A_i + B_i \leq 1$.*

(3) For every V triangle,

	T_i	U_i
$e_{i-1,i}$	$X_{i-1}([1 - A_i, 1])$	$X_{i-1}((1 - A_i, 1])$
$e_{i,i+1}$	$X_i([0, B_i])$	$X_i([0, B_i))$
r_i	$R_i([0, C_i])$	$R_i([0, C_i))$

Consequently

$$G_i = \begin{cases} X_i([B_i, 1 - A_{i+1}]), & B_i + A_{i+1} \leq 1, \\ \emptyset, & B_i + A_{i+1} > 1, \end{cases}$$

including a singleton in the equality case.

Solution.

Body consequence. Part 1 proves theorem 2.5. Part 2 proves theorems 2.7 and 2.10; part 3 proves theorems 2.8 and 2.9.

A.4. Module A.4: signed-center optimization.

Exact interface. Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and $N_E(T_C) \in \{1, 2\}$. A hexagon symmetry normalizes a positive trace to $e_{0,1}$. The module must determine its positive boundary traces and radial exits, including the point-contact boundary between CE1 and CE2.

Variables and feasible set. Use the affine chart

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

The orientation parameter and its complements are

$$\begin{aligned} 0 < R < 1, \quad W &:= 1 - R, \quad E := \sqrt{1 - RW}, \\ \eta &:= 1 - E, \quad P := E(1 - E). \end{aligned}$$

The center slacks satisfy $\alpha, \delta \geq 0$, and are strictly positive for the closure of an original open C triangle. Put

$$k := \eta + \alpha + \delta, \quad \Delta_R := P - \alpha - W\delta > 0, \quad \Delta_L := P - R\alpha - \delta.$$

Objective. Determine the exact right and left trace intervals, their sign classification and total length, all six radial exits, the exact midpoint set, and the CE1/CE2 boundary caps.

Exact result.

Proposition A.4 (Signed CE1/CE2 center normal form). *On the feasible set above, define the three side slacks by*

$$\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ (19) \quad F_1 &= Rb + Wa - k, \quad T_C = \{F_0, F_1, F_2 \geq 0\}. \\ F_2 &= W + \delta - b + Ra. \end{aligned}$$

Then

$$(20) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta \right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the only possible companion trace is

$$(21) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha \right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

In particular,

$$(22) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six exits, measured from O toward V_i , are

$$(23) \quad \begin{aligned} d_0^C &= E - \alpha - \delta, & d_1^C &= \frac{\delta}{R}, & d_2^C &= \delta, \\ d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, & d_4^C &= \alpha, & d_5^C &= \frac{\alpha}{W}. \end{aligned}$$

Moreover,

$$(24) \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

Solution. The remaining proof-used consequences belong to the same feasible set.

Lemma A.5 (CE2 total slack). *In the CE2 range, with $T := \alpha + \delta$,*

$$(25) \quad T < \frac{\eta}{E}.$$

Corollary A.6 (Common center-boundary formula). *The total C-triangle boundary contribution is*

$$(26) \quad L_{\partial H}(T_C) := \mathcal{H}^1(T_C \cap \partial H) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4},$$

whereas in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Lemma A.7 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, define*

$$M_0(q) := \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

If a unit equilateral triangle reaches parameters q, b on two adjacent boundary edges, then $b \leq M_0(q)$. The function $M_0(q)$ is strictly decreasing, and for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \quad 1 - M_0(q) > \frac{q}{2}.$$

Here the argument-free M_0 continues to denote the radial midpoint.

Body consequence. The unique-midpoint identity supplies theorem 2.13. The trace formula and its CE1/CE2 caps supply the compact center rows used by the trace-length interface. The exact endpoints, exits, total slack, and diameter transfer supply later appendix calculations without exposing signed parameters in the body.

A.5. Module A.5: strict-handoff feasibility.

Exact interface. Assume the six open V triangles cover ∂H . The body selector is

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1} < A_i, \quad b_i := \xi_i < B_i.$$

Variables and feasible set. Put

$$\ell_i := 1 - A_{i+1}, \quad u_i := B_i.$$

The exact incident open traces are $X_i([0, u_i))$ and $X_i((\ell_i, 1])$. Boundary coverage is therefore equivalent to the six strict overlap conditions $\ell_i < u_i$.

Objective. Choose one $\xi_i \in (\ell_i, u_i)$ on every edge, while preserving the unique actual supercritical index when $N_+ = 1$ and producing at least two selected supercritical indices when $N_+ \geq 2$.

Exact result.

Lemma A.8 (Strict-handoff selector). *The six selector intervals are nonempty. Every simultaneous choice produces $0 < a_i < A_i$, $0 < b_i < B_i$, and*

$$X_{i-1}(1 - a_i), X_i(b_i) \in U_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

If $N_+ = 1$, every choice has exactly one index with $a_i + b_i > 1$, equal to the actual supercritical index. If $N_+ \geq 2$, a simultaneous choice exists with at least two indices satisfying $a_i + b_i > 1$.

Solution.

Body consequence. This supplies every clause, including endpoint strictness, in theorem 2.11.

A.6. Module A.6: shared enclosure and local admissibility.

Exact interface. The enclosure number Λ , canonical anchor hull $K(a, b, c)$, admissible set $\mathcal{A}$, and radial envelope $c_{\max}$ are defined in theorems 5.1 and 5.3. This module evaluates those objects once for use by all later appendices.

Variables and feasible set. Let R be counterclockwise rotation through $2\pi/3$. For every nonempty compact $K \subset \mathbb{R}^2$ put

$$h_K(n) := \max_{x \in K} \langle x, n \rangle.$$

Translate the canonical hull's vertex V_0 to the origin and apply a rigid motion so that

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \quad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right).$$

Thus its generators become

$$O' = 0, \quad A = au, \quad B = bv, \quad C = c(u + v), \quad K = \text{conv}\{O', A, B, C\}.$$

For $a, b, c \in [0, 1]$ set

$$\begin{aligned} s &:= a + b, & d &:= a^2 + ab + b^2, & q &:= s^4 - s^2 + ab, \\ m &:= \min\{a, b\}, & M &:= \max\{a, b\}. \end{aligned}$$

and

$$\begin{aligned} F_L(a, b, c) &:= c^4 - c^2 + mc - m^2, \\ F_T(a, b, c) &:= (s^2 - 1)c^2 + Mc - M^2, \\ F_S(a, b, c) &:= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Objective. Evaluate $\Lambda(K)$ by support functions; determine the exact projection of $\mathcal{A}$ to the (a, b) -plane; and, on that projection, calculate the attained maximum $c_{\max}(a, b)$ with the geometric connected-component selectors retained.

Exact result.

Lemma A.9 (Shared equilateral-enclosure gauge). *For every nonempty compact K ,*

$$(27) \quad \Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^2 h_K(\mathbf{R}^j n).$$

If K is a polygon with at least two distinct points, a minimizing normal may be chosen perpendicular to a hull edge.

Lemma A.10 (Exact local admissible set and radial envelope). *The exact projection is*

$$(28) \quad \{(a, b) \in [0, 1]^2 : \exists c \in [0, 1] (a, b, c) \in \mathcal{A}\} = \{(a, b) \in [0, 1]^2 : a^2 + ab + b^2 \leq 1\}.$$

On $[0, 1]^3$, the admissible set is

$$(29) \quad \mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S,$$

where

$$\begin{aligned} \mathcal{A}_L &= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &= \{d \leq 1, s > 1, F_S \leq 0, c \leq 1/2\}. \end{aligned}$$

It is compact, coordinatewise downward closed, and invariant under exchange of a, b . Every c -fiber therefore has an attained maximum, given by

$$(30) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \leq 1, q \leq 0, (a, b) \neq (0, 0), \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & s \leq 1, q > 0, \\ \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

In particular, $c_{\max}(a, b) = c_{\max}(b, a)$.

Solution.

Body consequence. The support formula is the Appendix A supplier for theorem 5.1. The projection, component selectors, attainment, symmetry, and formula (30) supply theorem 5.3 and every later use of $c_{\max}$.

A.7. Module A.7: midpoint caliper optimization.

Exact interface. The geometric consequence required by theorem 2.12 is the implication

$$\begin{aligned} \{0, au, bv, (u + v)/2\} &\subseteq T, \\ T \text{ a closed unit equilateral triangle} &\implies a + b \leq 1. \end{aligned}$$

Variables and feasible set. Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N , put

$$(31) \quad W_S(N) := \sum_{k=0}^2 \max_{z \in S} \langle z, R^k N \rangle.$$

For the midpoint instance take

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \quad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right), \quad S = \left\{ 0, au, bv, \frac{u+v}{2} \right\}, \quad a, b \geq 0.$$

Objective. Reduce the least-equilateral-enclosure test to finitely many hull-edge normals and evaluate all four edge calipers when $a + b > 1$.

Exact result.

Theorem A.11 (Finite caliper certificate). *If a finite set S has at least two distinct points, then S lies in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ such that*

$$(32) \quad W_S(\varepsilon J(q - p)) \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

For $S_x = \{O, A, B, x\}$, every clause is polynomial of degree at most four in the coordinates of A, B, x , and at most $12 \cdot 4^3$ clauses are required. A singleton lies in equilateral triangles of arbitrarily small positive side.

Lemma A.12 (Four-point midpoint caliper). *If a closed unit equilateral triangle contains $0, au, bv, (u + v)/2$, then $a + b \leq 1$. If all four points have a positive containment margin, then $a + b < 1$.*

Solution.

Body consequence. The second result, with the local V-triangle chart transported by a hexagon symmetry, supplies theorem 2.12. The finite certificate remains available to every later finite-witness module.

APPENDIX B. TRACE-LENGTH OPTIMIZATION

This appendix solves the trace maximization and perimeter-budget problems whose compact outputs are used in Section 3. Each module starts from the exact body trace functional in Definition 3.1.

B.1. Skeleton-trace optimization.

Exact interface. The target is the exact trace functional L_S from Definition 3.1; the compact body consequence is Theorem 3.4.

Variables and feasible set. The C-triangle variables range over the signed CE1/CE2 domain of theorem A.4. A V triangle ranges over the closed unit equilateral triangles containing its distinguished vertex, with the displayed reach and radial-incidence restrictions in the lemmas below.

Objective. For each indicated class, maximize $L_S(T)$.

Exact result. On the four feasible classes below, the exact sufficient caps are, respectively,

$$L_S(T_C) < \frac{3}{2}, \quad L_S(T_i) < \frac{3}{2}, \quad L_S(T_i) \leq 2, \quad L_S(T_i) < \frac{3}{2}.$$

Solution. The following four lemmas give the complete calculations for those caps.

Lemma B.1 (Center skeleton cap). *If T_C is a CE1 or CE2 C triangle and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Lemma B.2 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle, so $V_i \in \text{int}(T_i)$. If*

$$\mathcal{H}^1(T_i \cap r_j) > 0 \quad \text{for some } j \in \{i-1, i+1\},$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Lemma B.3 (No-support skeleton cap). *If a V triangle has $A_i + B_i \leq 1$ and*

$$\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \leq 2.$$

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions u, v meeting at 120° , suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u+v),$$

with $0 \leq a \leq b \leq 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$(33) \quad \begin{aligned} a^2 + ab + b^2 &\leq 1, & c &\leq \frac{1}{2}, \\ (a^2 - 1)c^2 + (2ab^2 + b)c + (b^4 - b^2) &\leq 0, \end{aligned}$$

where c lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in a and, at $a = 1 - b$, equals $b^2(2b - 1)^2$; it is therefore positive when $a + b > 1$. Hence $c \leq 1/2$ selects its smaller-root component. Formula (33) follows by fixing in the support sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

Lemma B.4 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Body consequence. These four optima supply the four skeleton rows used in Theorem 3.4.

B.2. Vd1/Vd2 corner optimization.

Exact interface. The public outputs are the Vd1/Vd2 boundary row of Theorem 3.2 and the branch in Proposition 3.5.

Variables and feasible set. Let T be the closure of an original Vd1 or Vd2 triangle at V_0 . Use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Objective. Determine the exact half-plane family and maximize the boundary sum, first without and then with an adjacent-midpoint containment constraint.

Exact result. The exact half-plane and ray descriptions are (B.5) and (B.5). Their optimized consequences are

$$A + B < \frac{1}{2}, \quad M_{i-1} \in T_i \text{ or } M_{i+1} \in T_i \implies A_i + B_i < \frac{1}{3}.$$

Solution.

Proposition B.5 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 triangle at V_0 , and let A, B be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$T = \left\{ (x, y) : \begin{array}{l} x - (t+1)y \leq A, \\ ty - (t+1)x \leq tB, \\ tx + y \leq d - A - tB \end{array} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{l|l} r_0 & 0 \leq s \leq \frac{d - A - tB}{t+1} \\ r_1 & \frac{t(1-B)}{t+1} \leq s \leq \frac{d - A - tB - 1}{t} \\ r_5 & \frac{1-A}{t+1} \leq s \leq d - A - tB - t. \end{array}$$

Intersecting with $[0, 1]$ gives the actual traces. Moreover,

$$(34) \quad A + B < \frac{1}{2}.$$

Lemma B.6 (Vd2 adjacent-midpoint cap). *If an open role U_i has Vd2 closure $T_i = \overline{U_i}$ and T_i contains M_{i-1} or M_{i+1} , then its exact boundary reaches satisfy*

$$A_i + B_i < \frac{1}{3}.$$

Body consequence. The first result supplies the Vd1/Vd2 row of Theorem 3.2; the second supplies Proposition 3.5.

B.3. Boundary-trace optimization.

Exact interface. The objective is the body functional $L_{\partial H}$, and the public result is Theorem 3.2.

Variables and feasible set. The center ranges over the three exact classes determined by $N_E(T_C)$. Each V triangle ranges over its exact normalized V type and its actual maximal reaches (A_i, B_i) .

Objective. Maximize $L_{\partial H}$ separately on every row of the public table.

Exact result. The optimum in each feasible class is the corresponding entry of (3).

Solution. The signed center normal form gives both positive center traces from one formula, and the remaining rows follow from the exact V-triangle constraints.

Lemma B.7 (Boundary-trace table calculation). *For closures of original open triangles, the following bounds hold:*

<i>triangle</i>	<i>hypothesis</i>	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
T_C	CE2	$< \frac{1}{2}$
T_i	Vd0, $A_i + B_i \leq 1$	≤ 1
T_i	Vd0, $A_i + B_i > 1$	$\leq \frac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$< \frac{1}{2}$
T_i	<i>T3-like</i>	< 1 .

Body consequence. The seven exact maxima are the rows of Theorem 3.2.

B.4. Perimeter-budget optimization.

Exact interface. The center and V-triangle boundary contributions are those of Theorem 3.2. The public routing consequence is Proposition 3.6.

Variables and feasible set. The signed center variables range over the CE1 or CE2 cells of theorem A.4. The remaining variables are the numbers and strict caps of the classified V triangles in the stated routing branch.

Objective. Maximize the total perimeter trace subject to those caps and compare it with $\mathcal{H}^1(\partial H) = 6$.

Exact result. The exact deficit, small-slack, endpoint-slack, and routing conclusions are the statements of the four results below.

Solution. Their proofs perform the signed substitutions and strict endpoint checks.

Lemma B.8 (Master perimeter deficit). *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has a center contribution L_C , n supercritical Vd0 V triangles, $m \geq 1$ distinguished nonsupercritical V triangles with strict boundary caps $q_1, \dots, q_m < 1$, and $6 - n - m$ remaining V triangles whose boundary contributions are at most 1. If

$$(35) \quad L_C + n\theta \leq \sum_{j=1}^m (1 - q_j),$$

then the seven triangles do not cover ∂H .

Lemma B.9 (Small center slacks in the one-Vd1/Vd2 branch). *Assume there is exactly one supercritical Vd0 V triangle, exactly one Vd1/Vd2 V triangle, and four nonsupercritical Vd0 V triangles. Then every candidate surviving the perimeter budget is CE2 and satisfies*

$$(36) \quad \boxed{\alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.}$$

Lemma B.10 (CE2 initial endpoints dominate the total slack). *Assume $\Delta_R > 0$ and $\Delta_L > 0$. Put*

$$T = \alpha + \delta, \quad x = \frac{\eta + T}{W}, \quad y = \frac{\eta + T}{R}.$$

Then

$$(37) \quad \boxed{T < \frac{1}{2} \min\{x, y\}.$$

Corollary B.11 (Length branches closed by the common budgets). *The following branches are impossible.*

- (1) $N_{\text{gap}} = 0$, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;
- (2) $N_+ = 0$ and at least one Vd1/Vd2 V triangle, for either CE1 or CE2;
- (3) CE1, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;
- (4) CE2, $N_+ = 1$, and at least two Vd1/Vd2 V triangles;
- (5) CE2, $N_+ = 1$, with a Vd2 triangle T_i containing M_{i-1} or M_{i+1} ;
- (6) CE1 or CE2 with $N_+ + N_{\text{sp}} \geq 3$;
- (7) CE1 or CE2 with $N_+ \geq 2$.

Lemma B.12 (Exact trace-routing arithmetic). *Every configuration satisfying one of the six hypotheses in Proposition 3.6 has total available trace strictly smaller than the target trace required in that proposition.*

Body consequence. The preceding solved budget problem supplies Proposition 3.6, including its center-independent $N_{\text{gap}} = 0$ rows.

APPENDIX C. AREA-LOSS OPTIMIZATION

This appendix solves the local and cyclic variational problems defined in Section 4. The strict handoff selector is the one fixed in Proposition 2.11; it is not redefined here.

C.1. Local square-loss optimization.

Exact interface. The feasible family $\mathcal{T}(a, b)$ and objective $\mathcal{L}$ are those of Definition 4.1; the public output is Theorem 4.2.

Variables and feasible set. Fix $(a, b) \in [0, 1]^2$ with $\mathcal{T}(a, b) \neq \emptyset$. Normalize the distinguished vertex to V_0 and range over every orientation and translation of a closed unit equilateral triangle in $\mathcal{T}(a, b)$.

Objective. Minimize $\mathcal{L}(T)$ over $T \in \mathcal{T}(a, b)$.

Exact result. The optimized lower bounds are (4) and (5).

Solution.

Lemma C.1 (Local wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

Lemma C.2 (Solved local square-loss problem). *For every feasible pair $0 \leq a, b \leq 1$ and every realizing closed unit equilateral triangle T ,*

$$(38) \quad \mathcal{L}(T) \geq \min\{a, b\}^2.$$

If $a + b > 1$, then

$$(39) \quad \mathcal{L}(T) \geq \max\{a, b\}^2.$$

Consequently, taking the infimum over $T \in \mathcal{T}(a, b)$ gives the same inequalities for $\mathcal{L}(a, b)$.

Body consequence. The two optimized lower bounds are exactly Theorem 4.2.

C.2. T3-like area-loss optimization.

Exact interface. The input is an actual T3-like triangle and selected reach lower bounds as in Theorem 4.3.

Variables and feasible set. Normalize to V_0 . The triangle ranges only over the T3-like part of the closed unit-triangle family; in particular $(o, n) = (2, 1)$. The translation in Proposition 2.7 is used only as a closed-trace majorant.

Objective. Minimize the normalized loss of the actual T3-like triangle.

Exact result. The exact nonsupercritical and actual-triangle loss conclusions are (6) and (7).

Solution.

Lemma C.3 (Solved T3-like loss problem). *Let T_i be an actual T3-like V triangle, let $0 \leq a_i \leq A_i$ and $0 \leq b_i \leq B_i$, and let $0 \leq \mu \leq 1/2$. Then*

$$(40) \quad A_i + B_i \leq 1.$$

Moreover,

$$(41) \quad a_i, b_i \geq \mu \implies \mathcal{L}(T_i) \geq 2\mu - 4\mu^2.$$

Body consequence. This solved restricted optimization supplies the actual-triangle bound in Theorem 4.3; it makes no assertion about unrestricted $\mathcal{L}(a, b)$.

C.3. Cyclic area-loss optimization.

Exact interface. Use the strict handoff selector of Proposition 2.11 and the two hypotheses stated in Lemma 4.4.

Variables and feasible set. The variables are the cyclic selector tuple $(\xi_0, \dots, \xi_5)$ in the product of the six open selector intervals. The local losses range over the six actual V triangles realizing the selected lower bounds.

Objective. Minimize $\sum_{i=0}^5 \mathcal{L}(T_i)$ on each of the two selected-ascent classes.

Exact result. On each of the two feasible selector classes in Lemma 4.4, the exact conclusion is

$$\sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Solution.

Lemma C.4 (Solved cyclic area-loss problem). *Under either exact hypothesis in Lemma 4.4,*

$$\sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Body consequence. The two strict minima give Lemma 4.4; together with the structural selector they yield Proposition 4.5 for every center class whenever $N_{\text{gap}} = 0$.

APPENDIX D. NONZERO-GAP FINITE-ENCLOSURE OPTIMIZATION

This appendix supplies the solved calculations used by the nonzero-gap finite-enclosure interfaces. The variational objects Λ , $\mathcal{A}$, $c_{\max}$, and $C_{\pm}$ remain those defined in the body; the shared enclosure gauge and the local admissible set are evaluated in Appendix A. Each calculation below has a separate label so that the body cites only the result it uses.

D.1. Module D.1: neighboring-ray capacity.

Exact interface. The variational capacities C_+ and C_- are the exact closed-triangle maxima of theorem 5.5; the anchor comparison uses $c_{\max}$ from theorem 5.3.

Variables and feasible set. Take $0 \leq a, b \leq 1$ with $a + b \leq 1$. The maximizing radial coordinate is $c \in [0, 1]$, with the closed containment and reflection convention fixed in theorem 5.5. For the nonlinear branches below, $0 \leq a \leq 1/2$ and the selected cubic root lies in $[a, 1]$.

For $0 \leq a \leq 1/2$, let $p_*(a)$ be the unique root in $[a, 1]$ of

$$p^3 - (a + 2)p^2 + 2(a + 1)p - 1 = 0,$$

and put

$$\sigma(a) := 1 - p_*(a), \quad \tau(a) := 1 - a - (p_*(a) - a)(1 - p_*(a)).$$

Objective. Determine the attained maximum neighboring-ray reach $C_+(a, b)$, its reflected value $C_-(a, b)$, and the exact comparison of either neighboring capacity with the own-role capacity $c_{\max}(a, b)$.

Exact result.

Proposition D.1 (Exact neighboring-ray formula). *For $0 \leq a, b \leq 1$ and $a + b \leq 1$, the maximum in Definition 5.5 is attained and*

$$C_+(a, b) = \begin{cases} 1 - b, & a > 1/2, \\ 1 - b, & 0 \leq a \leq 1/2, \ 0 \leq b \leq \sigma(a), \\ p_*(a), & 0 \leq a \leq 1/2, \ \sigma(a) \leq b \leq \tau(a), \\ a + \frac{1}{2} - \sqrt{(a + b)^2 - \frac{3}{4}}, & 0 \leq a \leq 1/2, \ \tau(a) < b \leq 1 - a. \end{cases}$$

At $b = \tau(a)$ the selected value is $p_*(a)$, and $C_-(a, b) = C_+(b, a)$.

Lemma D.2 (Common-pair domination). *If $p, q \geq 0$ and $p + q \leq 1$, then*

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

Solution.

Body consequence. The attained piecewise evaluation supplies theorem 5.5; the final comparison supplies the compact body statement theorem 5.8.

D.2. Module D.2: type-aware endpoint forcing.

Exact interface. Use exactly the selected reaches, source-conditioned neighboring terms, Γ_i , and $D_i = (1 - \Gamma_i)V_i$ of theorem 5.6.

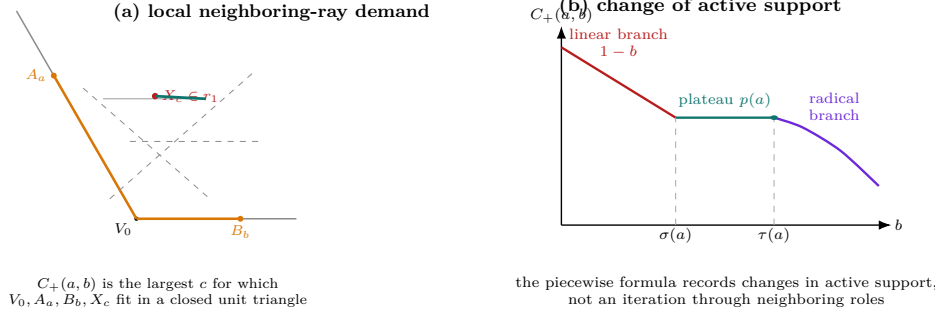


FIGURE 12. Geometry behind the neighboring-ray capacity $C_+(a, b)$. The local problem asks how far a triangle anchored at V_0 and the two boundary points A_a, B_b can reach on the neighboring arm r_1 . As b varies with a fixed, the active support pattern changes from the linear branch to a plateau and then to the radical branch, with transition points $\sigma(a)$ and $\tau(a)$. The graph is schematic; the transition order, endpoint selection, and branch labels are exact.

Variables and feasible set. The index is $i \in \mathbb{Z}/6\mathbb{Z}$; the selected reaches obey $0 \leq a_j \leq A_j$, $0 \leq b_j \leq B_j$, and $a_j^2 + a_j b_j + b_j^2 \leq 1$. A neighboring capacity is present only under the positive-trace implication displayed in the cited definition, which also places its pair in the domain $a + b \leq 1$.

Objective. Show that the largest admissible local closed radial endpoint still lies outside every open V role, so any seven-role cover must assign it to the center role. Exact result.

Lemma D.3 (Type-aware endpoint forcing). *The point D_i of Definition 5.6 satisfies*

$$D_i \in r_i \subseteq S, \quad D_i \notin \bigcup_{j=0}^5 U_j.$$

Consequently, a cover of S by the seven displayed roles forces $D_i \in U_C$.

Solution.

Body consequence. This endpoint-strict calculation is the unique Appendix D supplier for theorem 5.7.

D.3. Module D.3: nonsupercritical high-radial output.

Exact interface. Use the canonical admissible set $\mathcal{A}$ and actual reaches of theorem 5.3. The output function below is the attained maximum outgoing reach on its nonsupercritical fiber.

Variables and feasible set. Let $a, b, c \in [0, 1]$ satisfy $(a, b, c) \in \mathcal{A}$ and $a + b \leq 1$, with $c > 1/2$ for the high-radial branches. In the threshold form put $c = 1 - e$ with $0 < e < 1 - \sqrt{3}/2$; closed equality is allowed in the capacity, whereas the hypothesis on the actual incoming reach is strict.

For a nonsupercritical V triangle with incoming reach at least a and own radial reach at least c , define the attained outgoing capacity

$$\mathcal{B}_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A} \wedge a + b \leq 1\}.$$

For $0 < e < 1 - \sqrt{3}/2$, put

$$\epsilon(e) := \frac{1-e}{2} \left(1 - \sqrt{4(1-e)^2 - 3} \right).$$

Set

$$\mu := \frac{2\sqrt{3}-3}{4}.$$

The selected low root obeys

$$\begin{aligned} e &< \epsilon(e), \\ 0 < e \leq \mu &\implies \epsilon(e) < \frac{2e}{1-2e}, \quad 0 < e \leq \frac{1}{8} \implies \epsilon(e) \leq 2e + 5e^2. \end{aligned}$$

For $1/2 < c \leq \sqrt{3}/2$, put

$$h(c) := \frac{2c}{1 + \sqrt{16c^2 - 3}},$$

and, for $\sqrt{3}/2 \leq c < 1$, put

$$\ell(c) := \frac{c}{2} \left(1 - \sqrt{4c^2 - 3} \right), \quad r(c) := c - \ell(c).$$

Finally set

$$\begin{aligned} Q_-(a, c) &:= \frac{\sqrt{a^2 - ac + c^2}}{c} - a, \\ Q_+(a, c) &:= \frac{2(1-a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1-c^2)(1-a^2)c^2}}. \end{aligned}$$

Objective. Maximize the outgoing reach b on each high-radial nonsupercritical fiber, then determine the exact low-root threshold that forces the outgoing actual reach below the same root.

Exact result.

Proposition D.4 (Four direct high-radial outputs). *For $1/2 < c \leq \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1-a, & 0 \leq a \leq 1-c, \\ Q_+(a, c), & 1-c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1-a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1-a, & 0 \leq a \leq 1-c, \\ Q_+(a, c), & 1-c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1-a, & c \leq a \leq 1. \end{cases}$$

At $a = \ell(c)$ the selected value is $r(c)$; immediately to its right it is $\ell(c)$.

Lemma D.5 (Direct high-radial threshold). *Let a nonsupercritical V triangle have incoming, outgoing, and own-radial reaches A, B, C . If*

$$0 < e < 1 - \frac{\sqrt{3}}{2}, \quad C \geq 1 - e, \quad A > \epsilon(e),$$

then

$$B \leq \epsilon(e).$$

Solution.

Body consequence. The evaluated output catalogue and threshold supply the high-radial steps in theorems 5.14 and 5.15; later Appendix D modules cite these two exact results rather than re-evaluating the fiber.

D.4. Module D.4: strict-supercritical envelope.

Exact interface. The body defines M_c^{sup} as the supremum of outgoing reaches on the strict-supercritical part of $\mathcal{A}$ in the supported-tail interface preceding theorem 5.17. Variables and feasible set. Take $0 \leq c < 1/2$ and pairs $(a, b) \in [0, 1]^2$ satisfying $(a, b, c) \in \mathcal{A}$ and $a + b > 1$. For the relevant boundary reaches $1/2 \leq b < 1$, put

$$\kappa(b) := \frac{1 - b^2}{2 - b}.$$

The support-slack feasibility condition is the strict inequality $c < \kappa(b)$; it is not replaced by a blanket closure assertion for the whole face $a + b = 1$. At $c = 1/2$ use the body's declared continuous endpoint value.

Objective. Determine the supremal outgoing reach b , decide attainment on the strict fiber, and evaluate its one-sided endpoint limit.

Exact result.

Lemma D.6 (Strict-supercritical envelope). *For $0 \leq c < 1/2$ the variational quantity defined in the body is*

$$M_c^{\text{sup}} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

It is not attained by a strict-supercritical pair. Moreover

$$\lim_{c \uparrow 1/2} M_c^{\text{sup}} = \frac{1}{2} = M_{1/2}^{\text{sup}}.$$

Thus a strict-supercritical V triangle with own-radial reach at least c has outgoing boundary reach $B < M_c^{\text{sup}}$.

Solution.

Body consequence. This formula and its strict nonattainment supply the value M and the bound $B_1 < M$ in theorems 5.17 and 5.18.

D.5. Module D.5: selected-arc chord bounds.

Exact interface. Use the selected Q_+ component of the nonsupercritical output function from Module D.3 and the center-free boundary handoffs in theorem 5.15. Variables and feasible set. Let $0 < e < 1/2$, take $x \in [0, 1]$, and set $q = e + (1 - e)x$. The incoming coordinate p is the smaller-root point on the selected Q_+ component. For $e < 1 - \sqrt{3}/2$ it is the high-radial arc; for $1 - \sqrt{3}/2 \leq e < 1/4$ the CE1 use is the low-radial arc. These ranges keep every occurrence of $\epsilon(e)$ inside its declared domain.

Objective. Parametrize the selected output arc and obtain affine lower bounds for its outgoing reach along the two intervals used by the CE1 reverse path.

Exact result.

Lemma D.7 (Selected-arc chord bounds). *Let $0 < e < 1/2$. On the selected Q_+ component for radial demand $1 - e$, let p be the incoming reach and let q be the next incoming reach forced across a center-free edge. There is $x \in [0, 1]$ with*

$$q = e + (1 - e)x, \quad p = e + (1 - e)x - 1 + \sqrt{1 - x + x^2},$$

and hence

$$q - p = 1 - \sqrt{1 - x + x^2}.$$

This selected arc is increasing and concave. On each interval used by the CE1 reverse path,

$$q \geq p + \lambda(p - e),$$

with the following fully quantified choices. On the high-radial arc $0 < e < 1 - \sqrt{3}/2$, take

$$\lambda_{\text{hi}}(e) := \frac{e}{\epsilon(e) - e};$$

then

$$0 < e \leq \mu \implies \lambda_{\text{hi}}(e) > 1 - 4e, \quad 0 < e < 1 - \frac{\sqrt{3}}{2} \implies \lambda_{\text{hi}}(e) > 1 - 5e.$$

On the low-radial arc $1 - \sqrt{3}/2 \leq e < 1/4$, put

$$t_e := h(1 - e), \quad \lambda_{\text{lo}}(e) := \frac{1 - 2t_e}{t_e - e};$$

then

$$\lambda_{\text{lo}}(e) > \frac{1}{3} > 1 - 5e.$$

Solution.

Body consequence. The two strict chord estimates supply the successive return bounds in the Appendix D certificate for theorem 5.15.

D.6. Module D.6: Vd corner margins.

Exact interface. Use the Vd1/Vd2 corner-form actual reaches and the radial separation objects of theorem 5.21; selected lower bounds remain distinguished from the actual reaches A, B .

Variables and feasible set. Take $0 < A \leq 1$ and $0 \leq B \leq 1$ from the actual corner traces and $\theta > 0$, with $\Delta_\theta = \sqrt{\theta^2 + \theta + 1}$. The endpoint u_+ is used only when $\mathcal{H}^1(T_0 \cap r_1) > 0$. When selected demands are used, write them in lowercase and require $A \geq a_*$ and $B \geq b_*$.

Objective. Evaluate the corner's own-radial reach and far supported endpoint, and prove the strict margins needed to place the residual radial point beyond both.

Exact result.

Lemma D.8 (Vd corner margins). *In the Vd1/Vd2 corner form, let the actual boundary reaches satisfy $0 < A \leq 1$ and $0 \leq B \leq 1$, let $\theta > 0$, and put $\Delta_\theta = \sqrt{\theta^2 + \theta + 1}$. The own-radial reach is*

$$C_0 = \frac{\Delta_\theta - A - \theta B}{\theta + 1}.$$

If $\mathcal{H}^1(T_0 \cap r_1) > 0$, the far endpoint of that positive forward support is

$$u_+ = \frac{\Delta_\theta - A - \theta B - 1}{\theta}.$$

Consequently

$$C_0 < 1 - \min\{A, B\}, \quad u_+ < 1 - B \quad \text{when } \mathcal{H}^1(T_0 \cap r_1) > 0.$$

If $A \geq a_$ and $B \geq b_*$, then*

$$C_0 < 1 - \min\{a_*, b_*\}, \quad u_+ < 1 - b_* \quad \text{when } \mathcal{H}^1(T_0 \cap r_1) > 0.$$

Reflection supplies the backward-support version.

Solution.

Body consequence. The strict corner margins are used by the nonadjacent branch of theorem 5.21.

D.7. Module D.7: disk-finite-set calipers.

Exact interface. The enclosure objective is $\Lambda(\mathcal{D}_\eta \cup P)$ with Λ and the closed centered disk defined in theorem 5.1; here P is the displayed finite set.

Variables and feasible set. Let $\eta \geq 0$, let $\mathcal{D}_\eta = \mathbb{B}(O, \eta)$, and take $P = \{p_1, \dots, p_m\}$. The feasible orientation is a unit normal on the full normal circle; point-disk ties require $\|p_i\| \geq \eta$, and point-point ties require $p_i \neq p_k$.

Objective. Minimize the three-direction support sum for $\mathcal{D}_\eta \cup P$ and reduce every non-disk-only minimizer to a finite list of tie normals.

Exact result.

Proposition D.9 (Disk-finite-set calipers). *Let $\mathcal{D}_\eta$ be centered at O , let $P = \{p_1, \dots, p_m\}$, and let J denote rotation by $\pi/2$. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{J(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\eta p_i \pm \sqrt{\|p_i\|^2 - \eta^2} J p_i}{\|p_i\|^2} \quad (\|p_i\| \geq \eta).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

Solution.

Body consequence. This finite caliper reduction supplies the disk-plus-point evaluation used by theorem 5.9.

D.8. Module D.8: half-edge one-third radial envelope.

Exact interface. Use the anchor admissible set and attained radial envelope $c_{\max}$ from theorem 5.3.

Variables and feasible set. The ordered boundary demands satisfy $M \geq 1/2$, $0 < m \leq M$, and $M + m < 1$. The candidate radial coordinate is in $[0, 1]$ and is selected from the L or T cell of theorem A.10.

Objective. Give a strict upper bound for the maximum feasible own-radial reach in terms of the smaller boundary demand.

Exact result.

Lemma D.10 (Half-edge one-third radial envelope). *If*

$$M \geq \frac{1}{2}, \quad 0 < m \leq M, \quad M + m < 1,$$

then

$$c_{\max}(M, m) < 1 - \frac{m}{3}.$$

Solution.

Body consequence. The strict one-third bound supplies the local endpoint estimate in the adjacent branch of theorem 5.21.

D.9. Module D.9: rescuer endpoint and boundary-tail budget.

Exact interface. Use the supported interval, actual reaches, strict-supercritical envelope, and four-edge nonsupercritical path in theorems 5.17 and 5.18, together with the exact signed center traces and exits of theorem A.4.

Variables and feasible set. The special role has boundary endpoint a and radial interval $[c, u] \subset r_1$ with $\varepsilon = 1 - u$ and $M = M_c^{\sup}$. The center role is CE1 or CE2 in signed normal form, with $0 < R < 1$, $W = 1 - R$, $E = \sqrt{1 - RW}$, $\eta = 1 - E$, positive slacks α, δ , and $k = \eta + \alpha + \delta$. The four displayed adapter inequalities are assumed, T_1 is uniquely supercritical with radial reach at least c , and $T_2, \dots, T_5$ form the center-free nonsupercritical path.

Objective. Lower-bound the far boundary demand by M in both center-trace alternatives and test it against the four nonsupercritical boundary budgets.

Exact result.

Lemma D.11 (Rescuer endpoint and boundary-tail budget). *Let T_C be a CE1 or CE2 center role in the signed normal form of theorem A.4. Thus*

$$0 < R < 1, \quad W = 1 - R, \quad E = \sqrt{1 - RW}, \quad \eta = 1 - E, \quad \alpha, \delta > 0, \quad k = \eta + \alpha + \delta,$$

when its positive companion boundary trace on $e_{5,0}$ is present, that trace is $[k/W, R + \alpha]$; it may instead be empty or a point contact. Its exit on r_1 , measured from O , is δ/R . Suppose a special role at V_0 has boundary endpoint a on $e_{5,0}$ and supported interval $[c, u] \subset r_1$, measured from V_1 toward O . Put

$$\varepsilon = 1 - u, \quad M = M_c^{\sup}.$$

Assume

$$\begin{aligned} \varepsilon V_1 &\in U_C, & a + \varepsilon &\leq 1, \\ a &\leq 1 - M, & \frac{a}{a + \varepsilon} &\leq 1 - M. \end{aligned}$$

Assume also

$$U_C \cap \bigcup_{i=1}^4 e_{i,i+1} = \emptyset.$$

If T_1 is uniquely supercritical with own-radial reach at least c and T_2, T_3, T_4, T_5 are nonsupercritical along this four-edge path, then the skeleton is not covered.

Solution.

Body consequence. Together with the two branch-specific adapters, this common terminal proves theorems 5.17 and 5.18.

D.10. Module D.10: disk-plus-gap enclosure.

Exact interface. Use the enclosure number and centered disk of theorem 5.1, together with the exact gap interval and anchor capacity in theorem 5.9.

Variables and feasible set. For the auxiliary caliper take $\eta \geq 0$ and a point G with $\|G\| = \rho$. For the gap problem take $0 \leq p, q \leq 1$, $p + q \leq 1$, $c_* = c_{\max}(p, q)$, $m = \min\{p, q\}$, and the closed interval $J(p, q) = X_0([q, 1 - p])$; the endpoint case $p + q = 1$ remains feasible.

Objective. Evaluate the minimum enclosing side for a centered disk plus one point, then use that value to lower-bound the enclosure number of the disk-plus-gap set.

Exact result.

Lemma D.12 (Disk plus one point). *Let $\mathbb{B}(O, \eta)$ be the centered closed disk and let $\|G\| = \rho$. Then*

$$\Lambda(\mathbb{B}(O, \eta) \cup \{G\}) = \begin{cases} 2\sqrt{3}\eta, & \rho \leq 2\eta, \\ \frac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{\sqrt{3}}, & \rho > 2\eta. \end{cases}$$

Lemma D.13 (Complementary-gap calculation). *For $0 \leq p, q \leq 1$ with $p + q \leq 1$, set*

$$c_* := c_{\max}(p, q), \quad m := \min\{p, q\}, \quad d_* := 1 - c_*, \\ J(p, q) := X_0([q, 1 - p]), \quad \mathcal{D} := \mathbb{B}\left(O, \frac{\sqrt{3}}{2}d_*\right).$$

Then

$$\Lambda(\mathcal{D} \cup J(p, q)) \geq 1.$$

Under the common-pair hypotheses of the body, the exact same set is contained in U_C .

Solution.

Body consequence. The disk-plus-gap bound, including the singleton endpoint, supplies theorem 5.9 and the enclosure conclusion of theorem 5.10.

D.11. Module D.11: CE2 short-ray optimization.

Exact interface. Use the signed CE2 center variables of theorem A.4 and the exact body points D_2, D_4 from theorem 5.11.

Variables and feasible set. The selected boundary pair is $p = W - \alpha$, $q = R - \delta$ with $e = \min\{\alpha, \delta\}$ and $M = \max\{R, W\}$, subject to the strict CE2 inequalities in the signed normal form. The local radial optimization uses the compact fiber defining $c_{\max}(p, q)$.

Objective. Prove the strict capacity estimate $c_{\max}(p, q) < 1 - e$ and identify which of the two opposite body points lies beyond the corresponding C-triangle exit.

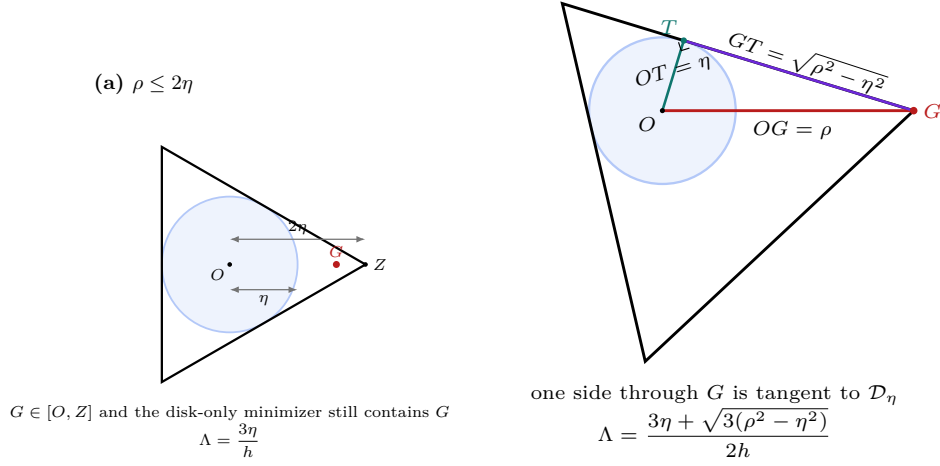
(b) $\rho > 2\eta$ (a) $\rho \leq 2\eta$ 

FIGURE 13. The two minimizing configurations in the disk-plus-point formula. For $\rho \leq 2\eta$, the centered equilateral triangle tangent to $\mathcal{D}_\eta$ already contains G . For $\rho > 2\eta$, the point G becomes a vertex of the minimizing triangle and one incident side is tangent to the disk. The tangent length $GT = \sqrt{\rho^2 - \eta^2}$ is the geometric source of the radical term.

Exact result.

Lemma D.14 (CE2 two-gap short-ray calculation). *Use the signed CE2 variables of theorem A.4 and put*

$$p := W - \alpha, \quad q := R - \delta, \quad e := \min\{\alpha, \delta\}, \quad M := \max\{R, W\}.$$

Then

$$0 \leq p, q \leq 1, \quad p + q \leq 1, \quad c_{\max}(p, q) < 1 - e.$$

Consequently the body points D_2, D_4 cannot both lie in T_C .

Solution.

Body consequence. The strict estimate and signed-exit alternative supply theorem 5.11.

D.12. Module D.12: transverse seven-point exclusion.

Exact interface. Use the exact seven-point set K_{tr} and the one-gap all-Vd0 normalization preceding theorem 5.13. The CE1 and CE2 center variables are those of theorem A.4; the boundary squeeze uses theorem A.7.

Variables and feasible set. For either signed center type put $X = R - \delta$ and $Q = (\eta + \alpha + \delta)/(2R)$. The five center-free handoffs and the retained radial points are exactly those required by theorems 5.14 and 5.15; no point on r_0 is part of the feasible witness.

Objective. Upper-bound the unique supercritical reach by $1 - Q$ from the actual gap endpoints, then rule out both possible signed center types for the same seven-point witness.

Exact result.

Lemma D.15 (Current transverse endpoint squeeze). *In the one-gap all-Vd0 normalization, suppose an open CE1 or CE2 unit triangle contains K_{tr} . Give that C triangle the signed variables of theorem A.4 and put*

$$X := R - \delta, \quad Q := \frac{\eta + \alpha + \delta}{2R}.$$

Then the actual reach of the unique supercritical role satisfies

$$A_0 < 1 - Q.$$

No radial point on r_0 is used.

Lemma D.16 (CE2 transverse threshold terminal). *No open CE2 unit equilateral triangle contains K_{tr} .*

Lemma D.17 (CE1 transverse return terminal). *No open CE1 unit equilateral triangle contains K_{tr} .*

Solution.

Body consequence. The endpoint squeeze and the two signed exclusions supply theorems 5.14 and 5.15; together with the body's forcing lemma they prove theorem 5.16.

D.13. Module D.13: T3-like supported-tail adapter.

Exact interface. Use the T3-like supported interval, actual boundary reach, and four adapter inequalities displayed in theorem 5.17.

Variables and feasible set. The original T3-like role has actual boundary reach A_0 and radial trace $[c, u]$. Its closed-trace majorant has the same A_0 and u but a possibly smaller near endpoint $\hat{c} \leq c$. The majorant normal form has source parameter $1 \leq \omega \leq 2/\sqrt{3}$, with $\omega = D$ and $\varrho = R_{\text{loc}}$; its rationalized variables x, ϑ obey the exact interval displayed in the solution. The body fixes $0 \leq c \leq 1/2 \leq u < 1$ and $M = M_c^{\text{sup}}$.

Objective. Verify the four endpoint and envelope inequalities needed to invoke the common rescuer-tail budget for the T3-like branch.

Exact result.

Lemma D.18 (T3-like supported-tail adapter). *Under the hypotheses of theorem 5.17, all four adapter inequalities displayed there hold.*

Solution.

Body consequence. Together with theorem D.11, these four inequalities supply theorem 5.17.

D.14. Module D.14: Vd1 supported-tail adapter.

Exact interface. Use the Vd1 supported interval, actual boundary reaches, and four adapter inequalities displayed in theorem 5.18.

Variables and feasible set. The Vd1 corner parameter satisfies $\theta \geq 1$, with $\Delta_\theta = \sqrt{\theta^2 + \theta} + 1$. The body interval obeys $0 \leq c \leq 1/2 \leq u < 1$, $\varepsilon = 1 - u$, and $M = M_c^{\text{sup}}$; the midpoint inequality selects the feasible corner component used below.

Objective. Bound the actual boundary endpoint and its normalized ratio by $1 - M$, and verify the endpoint containment and unit boundary-sum inequality required by the common rescuer-tail budget.

Exact result.

Lemma D.19 (Vd1 supported-tail adapter). *Under the hypotheses of theorem 5.18, all four adapter inequalities displayed there hold; in fact $A_0 + \varepsilon < 1$.*

Solution.

Body consequence. Together with theorem D.11, these four inequalities supply theorem 5.18.

D.15. Module D.15: adjacent Vd radial separation.

Exact interface. Use the adjacent alternative $(\sigma, \tau) = (0, 1)$ in theorem 5.21, the exact residual operator theorem 5.20, and the radial gap interface theorem 5.19 at $i = 2$. The signed small-slack and diameter inputs are theorems A.7 and B.9.

Variables and feasible set. Normalize the CE2 traces as in the body. With the signed variables of theorem A.4, put $k = \eta + \alpha + \delta$ and

$$I_L = \left[\frac{k}{W}, R + \alpha \right], \quad I_R = \left[\frac{k}{R}, W + \delta \right],$$

$\rho_R = \mathcal{R}_{I_R}(B_0)$, and $\rho_L = \mathcal{R}_{I_L}(A_0) > 0$ in the reduced candidate. The center-free handoffs give $A_1 \geq \rho_R$, $B_1 \geq \rho_L$, and $B_2 \geq \rho_L$.

Objective. Prove the strict separation margin $c_{\text{loc}} < 1 - d_C$ by comparing the left residual with the C exit and the adjacent Vd0 radial envelope.

Exact result.

Lemma D.20 (Adjacent Vd radial separation). *In the adjacent subcase $(\sigma, \tau) = (0, 1)$ of theorem 5.21, the radial interface at $i = 2$ satisfies*

$$c_{\text{loc}} < 1 - d_C.$$

Solution.

Body consequence. This strict local/C separation supplies the adjacent alternative of theorem 5.21.

D.16. Module D.16: nonadjacent Vd radial separation.

Exact interface. Use the nonadjacent alternative $\tau \in \{2, 3, 4\}$ in theorem 5.21, including the intervals I_R, I_L , the residual operator theorem 5.20, and the body point $D_\tau = \rho V_\tau$. The endpoint comparison uses theorem A.7.

Variables and feasible set. Put $x = k/W$, $y = k/R$ and $\rho = \min\{\rho_R, \rho_L\}$ for the two positive residuals of the normalized CE2 trace intervals. The routed C exit is the attained maximum d_τ^C , the actual Vd own-radial reach is C_τ , and all other roles retain the body type restrictions. The residual handoffs to the Vd role are $A_\tau \geq \rho_R$ and $B_\tau \geq \rho_L$.

Objective. Establish strict inequalities on both sides of ρ and exclude every nonlocal unit-diameter role from the selected radial point.

Exact result.

Lemma D.21 (Nonadjacent Vd radial separation). *In the nonadjacent subcase of theorem 5.21,*

$$d_\tau^C < \rho < 1 - C_\tau,$$

and every nonlocal role misses $D_\tau = \rho V_\tau$.

Solution.

Body consequence. These two strict radial estimates and the nonlocal distance audit supply the nonadjacent alternative of theorem 5.21.

D.17. Module D.17: CE2 one-Vd placement audit.

Exact interface. The body interface is the implication in theorem 5.23:

$$N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad d = 1 \implies S \not\subseteq U_C \cup \bigcup_{i=0}^5 U_i.$$

Variables and feasible set. Use the actual reaches A_i, B_i , the normalized types $(o(T_i), n(T_i))$, and the exact counts

$$\begin{aligned} d &= |\{i : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\}|, \\ t &= |\{i : (o(T_i), n(T_i)) = (2, 1)\}|, \quad N_{\text{sp}} = d + t. \end{aligned}$$

All indices lie in $\mathbb{Z}/6\mathbb{Z}$. The only assumed containment in the feasibility test is the hypothetical open cover $S \subseteq U_C \cup \bigcup_i U_i$.

Objective. Exhaust the relative positions of the unique supercritical role, the unique Vd1/Vd2 role, and the center-covered midpoint, while retaining every center-free and radial-locality hypothesis needed by the terminal to which a placement is routed.

Exact result.

Lemma D.22 (CE2 one-Vd placement audit). *Assume*

$$N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad d = 1.$$

Under a hypothetical cover of S , if $t \geq 1$, then $N_+ + N_{\text{sp}} \geq 3$. Otherwise $t = 0$, and after a dihedral symmetry

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

There are unique indices

$$\{\sigma\} = \{i : A_i + B_i > 1\}, \quad \{\tau\} = \{i : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\},$$

with $\sigma \neq \tau$ and

$$(o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\} \quad (j \neq \tau).$$

Exactly one of the following alternatives holds.

- (1) $\sigma = 0$, with $\tau \in \{1, 5\}$ or $\tau \in \{2, 3, 4\}$; these are respectively the adjacent and nonadjacent hypotheses of theorem 5.21.
- (2) $\tau = 0$, $M_\sigma \in T_0$, and $\sigma \in \{1, 5\}$. After reflection take $\sigma = 1$. Type Vd2 satisfies the hypotheses of theorem 3.5; type Vd1 has

$$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1, \quad C_1 \geq c,$$

and satisfies the hypotheses of theorem 5.18.

- (3) $\sigma \neq 0$ and $\tau \neq 0$, with $M_\sigma \in T_\tau$ and $\tau - \sigma \equiv \pm 1 \pmod{6}$. Type Vd2 satisfies the hypotheses of theorem 3.5. In type Vd1, rotate and, if necessary, reflect so that $(\tau, \sigma) = (0, 1)$. Then

$$(o(T_0), n(T_0)) = (1, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1,$$

$$A_i + B_i \leq 1 \quad (i \neq 1), \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5),$$

$$U_C \cap e_{0,1} = \emptyset, \quad \mathcal{H}^1(T_j \cap r_k) = 0 \quad (2 \leq j \leq 5, k \in \{0, 1\}),$$

so every hypothesis of theorem 5.22 holds.

Consequently the hypothetical cover is impossible.

Solution.

Body consequence. This exact placement partition, including the Vd2 length exits, both radial separation positions, the supported Vd1 position, and the full two-chart input hypotheses, supplies theorem 5.23.

D.18. Module D.18: CE1 one-gap reverse-path certificate.

Exact interface. Use the exact seven-point witness and the five center-free handoffs of theorem 5.15. Only the retained radial reaches at T_4, T_3, T_2 enter; no radial point on r_0 is part of this interface.

Variables and feasible set. Take the signed CE1 variables $0 < R < 1$, $w = 1 - R$, $E = \sqrt{1 - Rw}$, $\eta = 1 - E$, and $P = E(1 - E)$. The positive slacks $A = \alpha$ and $D = \delta$ satisfy the two signed CE1 inequalities displayed below. The five V roles have the stated type, handoff, and three closed radial-demand constraints.

Objective. Starting from a hypothetical upper bound on the returning actual reach A_0 , propagate the selected high-radial output twice and force a final contradiction at T_1 .

Exact result.

Proposition D.23 (CE1 three-transverse return certificate). *Use the signed CE1 center variables*

$$\begin{aligned} 0 < R < 1, \quad w = 1 - R, \quad E = \sqrt{1 - Rw}, \\ \eta = 1 - E, \quad P = E(1 - E), \end{aligned}$$

and write

$$A = \alpha, \quad D = \delta.$$

Assume

$$A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P.$$

Put

$$X = R - D, \quad m = \frac{A}{R}, \quad h_0 = \frac{\eta + A + D}{2R}.$$

Let T_0 be the unique supercritical role and let $T_1, \dots, T_5$ be nonsupercritical Vd0 roles. Assume

$$\begin{aligned} A_1 &\geq X, \\ B_i + A_{i+1} &\geq 1 \quad (1 \leq i \leq 4), \quad B_5 + A_0 \geq 1, \end{aligned}$$

and

$$C_4 \geq 1 - A, \quad C_3 \geq 1 - m, \quad C_2 \geq 1 - D.$$

Then

$$\boxed{A_0 > 1 - h_0.}$$

Solution.

Body consequence. This strict return estimate is the long CE1 supplier used by theorem D.17, and hence proves the compact body statement theorem 5.15.

D.19. Module D.19: two-chart Vd1 replacement.

Exact interface. Use exactly the normalized input roles, output reaches, output open traces, and recomputed gap rank of theorem 5.22. The input and output gap ranks are distinct objects.

CE1 reverse path: from a deficient terminal to the final contradiction

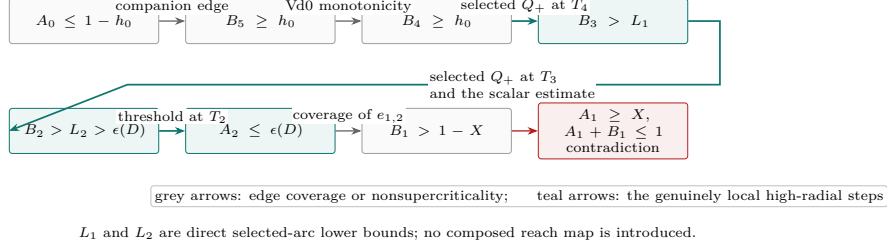


FIGURE 14. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the negation of the desired terminal bound at T_0 , moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

Variables and feasible set. The auxiliary admissible triple $(x, y, z) \in \mathcal{A}$ has $x \geq 1/2$ and $0 \leq z \leq 1/2$. For the replacement, the input is CE2 with $N_{\text{gap}} \in \{1, 2\}$, one supercritical role, the displayed Vd1 role, and the exact center-free trace hypotheses below. The chart parameters p_1, p_2, ε are selected by strict inequalities inside their nonempty open feasible intervals.

Objective. Construct distinct V_0 and V_1 replacement charts that preserve the five affected skeleton pieces with strict margins, make all six V roles nonsupercritical Vd0, and route from the recomputed output gap rank.

Exact result.

Lemma D.24 (Half-square admissibility). *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \geq \frac{1}{2}, \quad 0 \leq z \leq \frac{1}{2},$$

then

$$y \leq 1 - z.$$

Theorem D.25 (Two-chart replacement and recomputed-gap router). *Assume*

$$\begin{aligned} S &\subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \\ (o(T_0), n(T_0)) &= (1, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \quad A_i + B_i \leq 1 \quad (i \neq 1), \\ (o(T_i), n(T_i)) &\in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5), \quad U_C \cap e_{0,1} = \emptyset, \\ \mathcal{H}^1(T_j \cap r_k) &= 0 \quad (2 \leq j \leq 5, \quad k \in \{0, 1\}). \end{aligned}$$

Then there are two explicitly constructed open unit equilateral triangles U'_0, U'_1 , each containing its indexed vertex, for which all conclusions of theorem 5.22 hold. In particular, the output gap rank is recomputed from the output open traces and is not identified with the input gap rank.

Solution.

Body consequence. The half-square bound, both affine chart constructions, every strict preservation margin, and the three-way recomputed-gap routing supply theorem 5.22.

APPENDIX E. ZERO-GAP NINE-POINT OPTIMIZATION

This appendix supplies the exact coordinate evaluation of the public AB -union, the line-circle frontier points, their exclusions from the six V triangles, the Newton inner reduction, and the support-arc optimization used in the zero-gap body argument. Each solved module below states its public interface, feasible variables, objective, exact result, complete solution, and body consequence explicitly.

E.1. Strict AB -union and frontier module.

Exact interface. The body fixes

$$P_A = V_4 + a(V_3 - V_4), \quad P_B = V_4 + b(V_5 - V_4),$$

and the exact geometric family

$$\mathcal{R}_{AB}(a, b) = \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Variables and feasible set. Let e_A, e_B be unit vectors with angle 120° , put

$$O = 0, \quad A = ae_A, \quad B = be_B, \quad C = \{ue_A + ve_B : u, v \geq 0\},$$

identifying O, A, B with the images of V_4, P_A, P_B , respectively. In this chart the public feasible union is

$$(42) \quad \mathcal{R}_{AB}(a, b) = H \cap C \cap \bigcup \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ O, A, B \in T \end{array} \right\}.$$

The finite description in theorem A.11 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (42) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Objective. Determine the feasible union $\mathcal{R}_{AB}(a, b)$ exactly and identify the cyclic order of its two circular and two linear non-axis frontier pieces.

Exact result.

Theorem E.1 (Strict supercritical AB -union). *Assume*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned} \alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}. \end{aligned}$$

Then all four coefficients are positive and

$$(43) \quad \mathcal{R}_{AB}(a, b) = H \cap ((C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B)),$$

where

$$\begin{aligned} \mathcal{D}_A &= \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\}, \\ \mathcal{D}_B &= \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\}, \\ \mathcal{H}_A &= \{(u, v) : \alpha(u - a) + \beta v \leq 0\}, \\ \mathcal{H}_B &= \{(u, v) : \gamma u + \delta(v - b) \leq 0\}. \end{aligned}$$

Its non-axis frontier consists, in counterclockwise order, of the A -circle, the A -line, the B -line, and the B -circle.

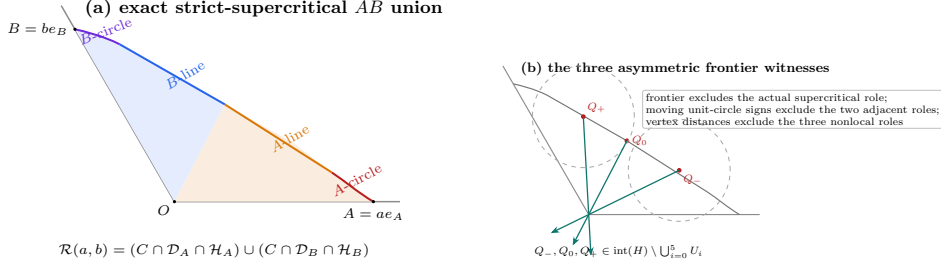


FIGURE 15. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A -circle arc, an A -line segment, a B -line segment, and a B -circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

Solution.

Body consequence. This supplies the exact feasible-union and maximal-line-piece evaluation used to define ℓ_-, ℓ_+ and Q_0 in the body, and is cited there as theorem E.1.

E.2. Handoff and nine-point frontier-forcing module.

Exact interface. The body interface is the exact chain and parameter domain in equations (8) to (10), the forced disk in equation (11), the three frontier points in equation (12), and the witness set in equation (13).

Variables and feasible set. Assume $N_{\text{gap}} = 0$, $N_+ = 1$, and $(d, t) = (0, 0)$; rotate the unique supercritical role to index 4 and use the strict handoffs ξ_i selected in the body. The feasible parameter pair is the strict component $0 < a, b < 1$, $a + b > 1$, $a^2 + ab + b^2 < 1$ determined below.

Objective. Determine the common radial envelope and the two line-circle roots, then prove that the resulting six radial points and three asymmetric frontier points are excluded from every V triangle.

Exact result. The exact outputs are supplied by theorems E.2 to E.5: they give the strict endpoint inequalities, the forced centered disk, the unique frontier roots, and $K_{\text{wit}}(a, b) \subset U_C$.

Solution.

Handoff stage.

Lemma E.2 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by theorem 2.11 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \quad b = \xi_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(44) \quad \begin{aligned} 0 < a, b < 1, \quad a = a_4 < A_4, \quad b = b_4 < B_4, \\ a + b > 1, \quad a^2 + ab + b^2 < 1, \end{aligned}$$

and every selected V triangle obeys

$$a_i \geq p, \quad b_i \geq q.$$

Radial stage. Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (44) gives

$$(45) \quad pq > (1 - s)(2 - s), \quad m > 1 - s, \quad s > 1 - m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of theorem A.10. On the present subcritical domain, equation (30) becomes

$$(46) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma E.3 (Direct radial forcing). *One has*

$$(47) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that every U_i is Vd0 and that $U_C, U_0, \dots, U_5$ cover H . Then, for every i ,

$$P_i^{\text{rad}} = (1 - c_*)V_i \in U_C.$$

Consequently

$$(48) \quad \mathbb{B}(0, r_*) = \{x : \|x\| \leq r_*\} \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_C, \quad r_* = h(1 - c_*).$$

Asymmetric frontier stage. We next define the three asymmetric witnesses. At V_4 use local coordinates

$$\Phi(u, v) := V_4 + u(V_5 - V_4) + v(V_3 - V_4), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

In this chart the two public adjacent-handoff circles have centers

$$\begin{aligned} X_2(b) &= \Phi(E_-), \quad E_- = (1 - b, 2 - b), \\ X_5(1 - a) &= \Phi(E_+), \quad E_+ = (2 - a, 1 - a), \end{aligned}$$

and hence

$$\mathcal{C}_- = \partial\mathbb{B}(\Phi(E_-), 1), \quad \mathcal{C}_+ = \partial\mathbb{B}(\Phi(E_+), 1).$$

The forward-reach lower bound is b and the backward-reach lower bound is a . Accordingly, set

$$\begin{aligned}\alpha &= \frac{h(b+2a-bD)}{2\rho}, & \beta &= \frac{h(b-a+(a+b)D)}{2\rho}, \\ \gamma &= \frac{h(-b+a+(a+b)D)}{2\rho}, & \delta &= \frac{h(2b+a-aD)}{2\rho},\end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of theorem E.1 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$(49) \quad J = \left(\frac{2b+a-aD}{D+3}, \frac{b+2a-bD}{D+3} \right).$$

Thus the two public line pieces have the coordinate evaluations

$$\ell_- = \{\Phi(L_-(\lambda)) : \lambda_* \leq \lambda \leq h^{-1}\}, \quad \ell_+ = \{\Phi(L_+(\mu)) : \mu_* \leq \mu \leq h^{-1}\}.$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1+t, t), \quad 1-a \leq t \leq b, \quad \text{and} \quad Z_3(y) = (y, 1+y), \quad 1-b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(50) \quad \begin{aligned}g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2.\end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned}\lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}.\end{aligned}$$

Lemma E.4 (Moving adjacent-triangle signs). *Throughout (44),*

$$(51) \quad \lambda_* < \lambda_- < h^{-1}, \quad \mu_* < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1-a, b]$,

$$(52) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1-b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Define the local and global witnesses by

$$(53) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and

$$Q_- := \Phi(Q_-^{\text{loc}}), \quad Q_0 := \Phi(Q_0^{\text{loc}}), \quad Q_+ := \Phi(Q_+^{\text{loc}}).$$

Thus J is the unique meeting point of the two maximal line pieces, and the first-root inequalities in theorem E.4 give exactly

$$\{Q_0\} = \ell_- \cap \ell_+, \quad \{Q_-\} = \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} = \ell_+ \cap \mathcal{C}_+.$$

Lemma E.5 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

The nine directly forced points are $P_0^{\text{rad}}, \dots, P_5^{\text{rad}}, Q_-, Q_0, Q_+$. Under a putative cover, theorems E.3 and E.5 therefore force the compact set

$$(54) \quad K_{\text{wit}}(a, b) = \mathbb{B}(0, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$

Body consequence. This is the full calculation supplying the body containments equations (11) to (13).

E.3. Newton reduction and four-overlap module.

Exact interface. Retain the public $K_{\text{wit}}(a, b)$ from equation (13) and the exact support cap

$$\text{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Variables and feasible set. Work on the strict AB domain in (44) and its nontrivial sheet $c_* > 2/3$. The Newton parameters are taken on the open segments from the line junction Q_0 to the first circle roots Q_- and Q_+ .

Objective. Replace the two independent first-root radicals by rational inner points and verify the ray order, radius bounds, and four consecutive support-cap intersections needed for the cyclic enclosure argument.

Exact result. The Newton containment is theorem E.6 and equation (55), and the complete four-overlap conclusion is theorem E.7.

Solution. For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (50), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g'_-(\xi_*)}, \quad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g'_+(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} P_- &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_- \right), \\ P_0 &= J, \\ P_+ &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+ \right), \end{aligned}$$

and convert them to global vectors.

Lemma E.6 (Newton inner reduction). *The points above have coordinates rational in a, b, D and satisfy*

$$P_- \in (Q_0, Q_-), \quad P_0 = Q_0, \quad P_+ \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \text{conv}(\mathbb{B}(0, r_*) \cup \{P_-, P_0, P_+\}),$$

and one has

$$(55) \quad \widehat{K} \subseteq \text{conv}(K_{\text{wit}}), \quad \Lambda(K_{\text{wit}}) \geq \Lambda(\widehat{K}).$$

Put

$$P_* := P_+ + \text{RP}_-.$$

For a disk $\mathbb{B}(X, r)$ let

$$\text{Cap}(X, r) := \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Ray order and overlap verification.

Proposition E.7 (Technical four-overlap verification). *Assume (44) and $c_* > 2/3$. Then*

$$\|P_-\|, \|P_+\| > r_*,$$

the rays $P_-, P_0, P_+, P_, \text{RP}_-$ occur in that cyclic order, and the four consecutive cap pairs*

$$\text{Cap}(P_-, 2r_*) \leftrightarrow \text{Cap}(P_0, 2r_*) \leftrightarrow \text{Cap}(P_+, 2r_*) \leftrightarrow \text{Cap}(P_*, r_*) \leftrightarrow \text{Cap}(\text{RP}_-, 2r_*)$$

overlap across their intervening short sectors.

Body consequence. These exact Newton and overlap data supply the nontrivial branch of the support-arc solution below, and hence the body conclusion theorem 5.25.

E.4. Support-arc enclosure module.

Exact interface. The body supplies the compact witness

$$K_{\text{wit}}(a, b) = \mathbb{B}(0, r_*) \cup \{Q_-, Q_0, Q_+\}$$

in equation (13), together with the exact level- h cap

$$\text{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Variables and feasible set. The parameters satisfy $0 < a, b < 1$, $a + b > 1$, and $a^2 + ab + b^2 < 1$. The disk branch is split exactly at $c_* = 2/3$; on the strict branch $c_* > 2/3$, use the Newton points and cap overlaps from the preceding module. All support directions range over the closed unit circle $\mathbb{S}^1$.

Objective. Determine the enclosure gauge of $K_{\text{wit}}(a, b)$ by proving the sharp sufficient bound $\Lambda(K_{\text{wit}}(a, b)) \geq 1$ on the entire feasible domain.

Exact result. The cyclic cap implication is theorem E.8; together with the disk branch and the four-overlap module it yields theorem E.9.

Solution. Let R denote counterclockwise rotation through $2\pi/3$. For $X \in \mathbb{R}^2$ and $r \geq 0$, recall

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\}, \quad \text{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Lemma E.8 (Cyclic support-arc covering). *Let K be compact and put*

$$\Sigma_K = K + RK + R^2K.$$

Suppose Σ_K contains the full R -orbits of

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*), \quad P_* := P_+ + RP_-.$$

Let θ_X denote the argument of the ray through X . Assume the rays

$$P_-, P_0, P_+, P_*, RP_-$$

occur in this cyclic order, every corresponding support arc is connected with half-width less than $\pi/2$, and, for each consecutive pair (X, r) , (Y, s) in

$$(P_-, 2r_*), (P_0, 2r_*), (P_+, 2r_*), (P_*, r_*), (RP_-, 2r_*),$$

one has

$$\text{Cap}(X, r) \cap \text{Cap}(Y, s) \cap \{n : \arg n \in [\theta_X, \theta_Y]\} \neq \emptyset,$$

where $[\theta_X, \theta_Y]$ is the intervening short angular interval. Then

$$\Lambda(K) \geq 1.$$

Theorem E.9 (Zero-gap nine-point optimization). *For every pair (a, b) in the strict domain*

$$0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

the exact public witness satisfies

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Body consequence. This solved optimization supplies theorem 5.25. Combined with $K_{\text{wit}}(a, b) \subset U_C$, it gives the compact body obstruction theorem 5.26 and its routing consequence theorem 5.27.

APPENDIX F. EXACT POLYNOMIAL POSITIVITY CERTIFICATE

The two mixed cap overlaps in the finite-enclosure method are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

F.1. Certificate manifest. The certificate has a machine-readable provenance manifest. The file `certificate_provenance.json`, in the numbered 3105X computation directory, identifies the six sparse-data shards and both exact verifiers by full Git blob and records their roles. They decode to one canonical sparse-polynomial transcript with SHA-256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

F.2. Rational radial upper envelopes. Retain the notation of appendix E. Thus

$$p = 1 - b, \quad q = 1 - a, \quad s = p + q, \quad m = \min\{p, q\}, \quad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \quad h = \frac{\sqrt{3}}{2}, \quad c_* = c_{\max}(p, q).$$

For the certificate formulas only, abbreviate

$$A := P_-, \quad C := P_+, \quad \eta := r_*, \quad I(X, r) := \text{Cap}(X, r).$$

The strict witness domain gives

$$(56) \quad mM > (1 - s)(2 - s), \quad m > 1 - s, \quad s > \frac{2}{3}.$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \quad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$(57) \quad \bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \quad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0).$$

Lemma F.1 (Branchwise radial upper bounds). *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \quad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \quad m = \frac{1 - U - z}{2}, \quad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\Xi(U, z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$

$$G(U, z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$

Thus

$$(58) \quad \bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \quad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}.$$

F.3. Gram reduction and the geometric implication. On the active cell let $\bar{c}$ denote the corresponding expression in (58), and put

$$\bar{\eta} = h(1 - \bar{c}), \quad e = 1 - \bar{c}, \quad t = 2\bar{c} - 1.$$

theorem F.1 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta}, \bar{\eta}, 2\bar{\eta}$.

Represent a global vector as (x, hy) and put $C' = R^{-1}C$. Define

$$N_A = \|A\|^2, \quad N_C = \|C\|^2, \quad \nu = \langle A, C' \rangle,$$

$$\Delta_0 = x_A y_{C'} - y_A x_{C'} > 0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$(59) \quad h^2 \Delta_0^2 = N_A N_C - \nu^2.$$

Define

$$(60) \quad Q_A = \Delta_0^2 - \|tA - eC'\|^2, \quad Q_C = \Delta_0^2 - \|tC' - eA\|^2.$$

Lemma F.2 (Residual signs imply the mixed cap intersections). *If $Q_A, Q_C \geq 0$, then*

$$I(C, 2\bar{\eta}) \cap I(C + RA, \bar{\eta}) \neq \emptyset,$$

and

$$I(C + RA, \bar{\eta}) \cap I(RA, 2\bar{\eta}) \neq \emptyset.$$

Consequently the same intersections hold with η in place of $\bar{\eta}$.

F.4. Eight exact integer-polynomial signs. The Newton witnesses A, C are rational functions of U, z, D . Substituting (58) into (60), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$(61) \quad \text{num}(Q_{B,X}) = 32(K+3)^2 \{A_{B,X}(U, z) + DB_{B,X}(U, z)\},$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C , both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

F.5. Three global positive-basis certificates. Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \quad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_\Delta = 1 - \frac{\sqrt{3}}{2}, \quad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

$$\begin{aligned} T : & \quad 0 \leq U \leq U_\Delta, \quad 0 \leq z^2 \leq \mathcal{P}(U), \\ L_0 : & \quad 0 \leq U \leq U_\Delta, \quad \mathcal{P}(U) \leq z^2 \leq \mathcal{G}(U), \\ L_1 : & \quad U_\Delta \leq U \leq U_{\max}, \quad 0 \leq z^2 \leq \mathcal{G}(U). \end{aligned}$$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \quad z = y \frac{9-x^4}{8x^2}, \quad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1,$$

and then $x = 1 + (\sqrt{3} - 1)r$. This maps $[0, 1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x , the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	(88, 22)	2047	2
$B_{T,A}$	(80, 20)	1701	0
$A_{T,C}$	(88, 22)	2047	2
$B_{T,C}$	(80, 20)	1701	0.

Every nonzero coefficient is strictly positive.

For an L -cell polynomial write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \quad w = z^2,$$

and

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}|O_F|$, hence $F \geq 0$ for $z \geq 0$. Use the exact charts

$$U = U_\Delta r, \quad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0 , and

$$U = U_\Delta + (U_{\max} - U_\Delta)r, \quad w = s\mathcal{G}(U)$$

for L_1 . The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r, s) = \sum a_{ij}r^i s^j$ has bidegree (m, n) , its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \leq k} \sum_{j \leq \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem F.3 (Exact mixed-overlap certificate). *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in (61) are nonnegative. Hence $Q_A, Q_C \geq 0$, and the two mixed cap intersections required by theorem E.7 hold.*

Remark F.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.